Exercise list — Topic 5: Nonverifiable income, strategic default, termination, debt structure

Worked solutions

Statements are in `Listas/lista-topic5-nonverifiable-income-termination.md` (and its PDF); each exercise is cited by number and printed page and is not restated. Short answers: `lista-topic5-nonverifiable-income-termination-solutions-short.pdf`. Editions are the local copies: Tirole, *The Theory of Corporate Finance* (2005); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001); Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Grinblatt–Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013).

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1. Control passing to creditors — Grinblatt & Titman 16.10, p. 593

What is asked. The exercise has two parts. *(preferences)*: explain how what shareholders want the firm to do differs from what creditors want, given the claims each one holds. *(decisions)*: explain which corporate decisions (risk, liquidation, investment, payouts, management) are likely to change once bankruptcy hands control of the firm to the debt holders.

Approach. Each claimant's preferences follow from the shape of its payoff as a function of firm value: equity is a call option on the firm (convex), debt is a short put plus a riskless bond (concave).

Whoever holds control maximises the expected value of *its own* claim, not total value; this is the debt holder–equity holder conflict of G&T §16.2. In the lecture's language, default is the event that transfers control to creditors, which is what makes their claim enforceable when income is not verifiable. A two-state numerical example makes each bias concrete.

Notation.

- $F > 0$: face value of the debt due now; $V_L \geq 0$: value if the firm is liquidated now.
- $\tilde{V}$: random value of the firm if it keeps operating (going-concern value), $E[\tilde{V}]$ its mean; risk neutrality and a zero interest rate, as in the chapter's examples.
- $E(\cdot)$, $D(\cdot)$: payoffs of equity and debt for a given total value V.

Step 1 — Two claims, two payoff shapes. By absolute priority, at a value V the claims pay

$$D(V) = \min(V, F), \tag{1}$$
$$E(V) = \max(V - F, 0). \tag{2}$$

Equity (2) is a call option on the firm: convex, it gains from upside and is protected on the downside. Debt (1) is concave: it gains nothing above F and bears the losses below F. Each group, when in control, picks the action that maximises the expected value of its own claim, not $E[\tilde{V}]$ (G&T Result 16.2, p. 562).

Step 2 — Equity holders' preferences while they control the firm. Near distress, (2) makes shareholders (and the managers who represent them) prefer: (i) *risk* — asset substitution, even into negative-NPV gambles, because the downside falls on creditors; (ii) *continuation* over liquidation — reluctance to liquidate, since liquidation pays them $\max(V_L - F, 0) = 0$ when $F > V_L$, like exercising an out-of-the-money option; (iii) *no new equity-financed investment* — debt overhang, since the existing debt captures much of the payoff; (iv) *quick payoffs* — shortsighted investment (G&T §16.2, pp. 562–578). Illustration with the book's Eastern Airlines numbers (p. 576): $F = 500$; continuing gives $\tilde{V} = 600$ or 200 with probability $\frac{1}{2}$ each; liquidating now gives $V_L = 480$ ($ millions). Using (2) and (1),

$$\begin{aligned} E_{\text{cont}} &= \tfrac{1}{2}\max(600 - 500, 0) + \tfrac{1}{2}\max(200 - 500, 0) && (3)\\ &= \tfrac{1}{2}(100) + \tfrac{1}{2}(0) = 50, \\ E_{\text{liq}} &= \max(480 - 500, 0) = 0, && (4)\\ D_{\text{cont}} &= \tfrac{1}{2}\min(600, 500) + \tfrac{1}{2}\min(200, 500) && (5)\\ &= \tfrac{1}{2}(500) + \tfrac{1}{2}(200) = 350, \\ D_{\text{liq}} &= \min(480, 500) = 480. && (6) \end{aligned}$$

Shareholders want to continue ($50 > 0$), creditors want to liquidate ($480 > 350$), and the firm is worth $E[\tilde{V}] = \frac{1}{2}(600) + \frac{1}{2}(200) = 400$ if it continues against 480 if it liquidates: equity control destroys 80.

Step 3 — Creditors' preferences once they take control. In bankruptcy (or after a covenant default) control passes to the debt holders or their representatives (G&T p. 576). By (1) they are *conservative*: they value the downside fully and the upside above F not at all. They therefore prefer (i) safe strategies and lower risk; (ii) liquidation or asset sales whenever V_L exceeds the expected value of their claim under continuation, which is efficient in Step 2's example; (iii) cash collection and cost cutting over long-horizon, risky investment. The bias can now run the other way: with $F = V_L = 500$ and $\tilde{V} = 800$ or 300 (own numbers), continuing is worth $E[\tilde{V}] = \frac{1}{2}(800) + \frac{1}{2}(300) = 550 > 500$, but

$$\begin{aligned} D_{\text{cont}} &= \tfrac{1}{2}\min(800, 500) + \tfrac{1}{2}\min(300, 500) && (7)\\ &= \tfrac{1}{2}(500) + \tfrac{1}{2}(300) = 400 < 500 = D_{\text{liq}}, \end{aligned}$$

so creditors liquidate a firm worth more alive (the Emruss case, pp. 577–578, has the same logic: senior creditors liquidate because liquidation pays them in full).

Answer (preferences). Equity holders (convex claim $\max(V-F,0)$) favour risk, continuation over liquidation, no new equity-financed investment and quick payoffs; debt holders (concave claim $\min(V,F)$) favour safety, liquidation whenever V_L exceeds the expected value of their claim under continuation, and cash collection. In the example, equity wants to continue ($50>0$), debt wants to liquidate ($480>350$).

Step 4 — How decisions change after bankruptcy. Relative to equity control: risk is cut, failing divisions are closed or sold, liquidation becomes more likely (sometimes too likely), dividends stop, and managers are often replaced. When Chapter 11 converts the debt into equity of the reorganised firm, the former creditors become residual claimants: debt overhang and the incentive to gamble disappear, and decisions move back toward maximising total value. Absolute-priority violations (p. 560) leave shareholders some bargaining power, so the shift of control is only partial.

Answer (decisions). After bankruptcy the firm takes less risk, sells or closes failing assets, liquidates more often (sometimes too often: with $\tilde{V}=800$ or 300 and $V_L=F=500$, creditors liquidate a firm worth $550>500$ alive), stops dividends and often replaces management; once the debt is converted into equity, decisions move back toward maximising total value.

Intuition. Each claimant behaves like the shape of its payoff: the convex claim gambles and delays liquidation, the concave claim plays safe and liquidates early. Shifting control to creditors in default is what makes their claim credible — the same role the termination threat plays in Topic 5 — but it replaces one bias by the opposite one unless the claims are restructured.

Common mistakes.

- Saying that creditors in control maximise firm value: they maximise $E[\min(\tilde{V},F)]$ and ignore the upside above F, so they can liquidate a firm worth more alive.
- Treating liquidation as always efficient (or always inefficient): it is efficient only when $V_L>E[\tilde{V}]$; the bias depends on who holds control.
- Forgetting absolute-priority violations and debt-for-equity swaps, which change the payoff shapes and hence the post-bankruptcy decisions.

Ref: Grinblatt & Titman Ch. 16, §16.1 "Bankruptcy", pp. 559–561; §16.2 "Debt Holder–Equity Holder Conflicts", pp. 561–578 ("The Reluctance to Liquidate Problem", pp. 575–578); Ch. 18, p. 643 (creditor influence makes managers "too conservative").

2. Duration of debt: proving (5.11) — Amaro de Matos 5.2.5 #3, p. 148

What is asked. Derive the face value r_1 of one-period (short-term) debt in Diamond's (1991) rating model, i.e. show that competitive lenders who must earn R per period set r_1 as in (5.11).

Approach. Short-term debt is priced by a one-period break-even condition: what the lenders expect to hold at $t=1$, averaged over the upgrade and downgrade states, must equal R per unit lent. The state probabilities come from the rating model ($f^u=1$: only good projects are upgraded), and the claim's $t=1$ value in each state is given in the text. This is the lecture's maturity question: short-term debt is repriced at $t=1$ on the basis of nonverifiable (but observable) news.

Notation.

- $f\in(0,1)$: date-0 probability that the project is good (credit rating); $f^u=1$, $f^d<f$: posterior after an upgrade / a downgrade.
- $e\in(0,1)$: probability that a good project is downgraded at $t=1$; bad projects are always downgraded (since $f^u=1$).
- r_1: face value of one-period debt due at $t=1$; V_s^u, V_s^d: its $t=1$ value after an upgrade / a downgrade.

- $R > 1$: gross one-period return of the lenders' alternative; L: liquidation value; X: payoff of a successful project; $\pi \in (0,1)$; $q^d = \pi + f^d(1-\pi)$: repayment probability after a downgrade (5.10).
- P_u, P_d: probabilities of an upgrade and a downgrade for a borrower of unknown type; $M \equiv \max(L, q^d X/R)$.

Step 1 — Probabilities of upgrade and downgrade. We need the weights of the two $t=1$ states. By the law of total probability over the two types, with $\Pr(\text{up} \mid \text{good}) = 1-e$ and $\Pr(\text{up} \mid \text{bad}) = 0$ (only good projects are upgraded, since $f^u = 1$):

$$\begin{aligned} P_u &= f\,(1-e) + (1-f)\,(0) \\ &= f(1-e), \qquad (8) \\ P_d &= 1 - P_u = 1 - (1-e)f. \qquad (9) \end{aligned}$$

The downgrade posterior is consistent with these by Bayes' rule, since a downgrade comes from a good project with probability fe and from a bad one with probability $(1-f)(1)$:

$$f^d = \frac{fe}{fe + (1-f)} = \frac{fe}{P_d}. \qquad (10)$$

Step 2 — Value of the claim at $t=1$. After an upgrade the borrower is known to be good and refinances r_1 with new debt of face value $r_1 R$, repaid with certainty, so

$$V_s^u = r_1. \qquad (11)$$

After a downgrade the value is the one stated in the text, $V_s^d = \min[r_1, \max(L, q^d X/R)] = \min(r_1, M)$.

Step 3 — Lenders' break-even. Lenders give $K=1$ at $t=0$ and require an expected value R at $t=1$ (one period at the per-period return R):

$$P_u V_s^u + P_d V_s^d = R. \qquad (12)$$

Substituting (8), (9) and (11) into (12):

$$\begin{aligned} \big(f(1-e)\big)(r_1) + \big(1-(1-e)f\big)V_s^d &= R, \\ (1-e)f\,r_1 &= R - [1-(1-e)f]V_s^d, \\ r_1 &= \frac{R - [1-(1-e)f]V_s^d}{(1-e)f}, \qquad (13) \end{aligned}$$

where the second line subtracts $[1-(1-e)f]V_s^d$ from both sides and the third divides by $(1-e)f > 0$. This is (5.11). ■

Step 4 — Solving the fixed point. Since V_s^d depends on r_1, (13) is an equation in r_1; we solve it case by case to see when the debt is risky. Case A, $r_1 \le M$: then $V_s^d = r_1$ and (13) gives

$$\begin{aligned} (1-e)f\,r_1 + [1-(1-e)f](r_1) &= R, \\ r_1\big[(1-e)f + 1 - (1-e)f\big] &= R, \\ r_1 &= R, \qquad (14) \end{aligned}$$

consistent if and only if $R \le M$: the debt is riskless. Case B, $R > M$: then $V_s^d = M$ and

$$r_1 = \frac{R - [1-(1-e)f](M)}{(1-e)f}. \qquad (15)$$

It is consistent ($r_1 > M$) because

$$r_1 - M = \frac{R - [1-(1-e)f]M - (1-e)f\,M}{(1-e)f} = \frac{R-M}{(1-e)f} > 0,$$

and $r_1 > R$ because $r_1 - R = [1-(1-e)f](R-M)/[(1-e)f] > 0$ (subtract $R = (1-e)fR/[(1-e)f]$ from (15)). The premium $r_1 - R$ pays for the shortfall $R - M$ that lenders bear in the downgrade state.

Answer (5.11). $r_1 = \dfrac{R - [1-(1-e)f]\,V_s^d}{(1-e)f}$, from the break-even $f(1-e)\,r_1 + [1-(1-e)f]\,V_s^d = R$. Since $V_s^d = \min(r_1, M)$, it gives $r_1 = R$ (riskless debt) when $R \le M$ and $r_1 = [R - (1-(1-e)f)M]/[(1-e)f] > R$ otherwise.

Intuition. Short-term lenders are paid in full whenever the borrower is upgraded and receive at most the refinancing or liquidation value after a downgrade; the face value is set so that the probability-weighted average of the two equals the safe return. Because the downgrade state carries the loss, r_1 rises with the probability $1-(1-e)f$ of a downgrade.

Common mistakes.

- Requiring R^2 instead of R: short-term debt runs one period, so the lenders' target at $t=1$ is R (long-term debt is the one priced against R^2).
- Writing the upgrade probability as f or $1-e$: it is $f(1-e)$, because bad projects are never upgraded ($f^u = 1$) and good ones are upgraded only with probability $1-e$.
- Treating V_s^d as a constant: it is $\min[r_1, \max(L, q^d X/R)]$, so (5.11) is a fixed point and must be checked for consistency.

Book answer. Matches the book's answer (p. 268), which writes the break-even condition $P_u V_s^u + P_d V_s^d = R$ with $P_u = f(1-e)$ and solves for r_1. Step 4 is an addition.

Ref: Amaro de Matos Ch. 5, §5.2.3 "The Choice of the Duration", pp. 139–141; Appendix C.5.2, p. 268.

3. Threat of termination: continuation probability — Amaro de Matos 5.2.5 #1, p. 148

What is asked. Show that the optimal liquidation probability $\beta_\circ$ after a declared zero cash flow, given by (5.8), falls when the probability q of the good cash flow rises (i.e. sign the derivative $\partial\beta_\circ/\partial q$).

Approach. With nonverifiable income, repayment is enforced only by the *threat of termination*: after a zero report the investor liquidates with probability $\beta_\circ$, set at the lowest level that lets her break even. The denominator of (5.8) is her expected recovery per unit of $\beta_\circ$, a q-weighted average of the recovery after a *strategic default* ($y/2$) and after a *liquidity default* ($L_\circ$). Differentiating the ratio and signing $y/2 - L_\circ$ answers the question.

Notation.

- $q \in (0,1)$: probability of the cash flow x at $t=1$; $y > 0$: second-period cash flow; $K > 0$: investment.
- $\beta_\circ \in [0,1]$: probability of liquidation after a declared zero cash flow (the continuation probability is $1-\beta_\circ$).
- $S = y/2$: entrepreneur's share in a strategic-default buyback, so the investor gets $y - S = y/2$.
- $\alpha \in [0,1]$: efficiency of an outside buyer; $c \sim U[0,\bar{c}]$: buyer's information cost, with $\alpha y/2 \le \bar{c}$; $L_\circ = \alpha^2 y^2/(4\bar{c})$: expected liquidation payoff.

- $\Pi(q)$: investor's expected recovery per unit of $\beta_\circ$, the denominator of (5.8).

Step 1 — Write $\beta_\circ$ as a ratio. From (5.8),

$$\beta_\circ(q) = \frac{K}{\Pi(q)}, \tag{16}$$

$$\Pi(q) = q\,\frac{y}{2} + (1-q)\frac{\alpha^2 y^2}{4\bar{c}}. \tag{17}$$

Step 2 — Differentiate. The sign of the comparative static is the sign of a derivative, so we compute it. Writing (16) as $\beta_\circ = K\,\Pi(q)^{-1}$ and applying the chain rule (K, y, α, $\bar{c}$ do not depend on q):

$$\begin{aligned}\frac{\partial \beta_\circ}{\partial q} &= K \cdot (-1)\,\Pi(q)^{-2} \cdot \Pi'(q) \\ &= -\frac{K\,\Pi'(q)}{\Pi(q)^2}.\end{aligned} \tag{18}$$

(17) is linear in q; $\frac{d}{dq}\left[q\,\frac{y}{2}\right] = \frac{y}{2}$ and $\frac{d}{dq}\left[(1-q)\frac{\alpha^2 y^2}{4\bar{c}}\right] = -\frac{\alpha^2 y^2}{4\bar{c}}$, so

$$\Pi'(q) = \frac{y}{2} - \frac{\alpha^2 y^2}{4\bar{c}}. \tag{19}$$

Substituting (19) and (17) into (18):

$$\frac{\partial \beta_\circ}{\partial q} = \frac{-K\left(\dfrac{y}{2} - \dfrac{\alpha^2 y^2}{4\bar{c}}\right)}{\left[q\,\dfrac{y}{2} + (1-q)\dfrac{\alpha^2 y^2}{4\bar{c}}\right]^2}. \tag{20}$$

Step 3 — Sign. The denominator of (20) is positive and $K > 0$, so the sign is the opposite of that of $\Pi'(q)$. Factor $y/2$ out of (19):

$$\begin{aligned}\Pi'(q) &= \frac{y}{2}\left[1 - \frac{\alpha^2 y^2}{4\bar{c}} \cdot \frac{2}{y}\right] \\ &= \frac{y}{2}\left[1 - \frac{\alpha^2 y}{2\bar{c}}\right] \\ &= \frac{y}{2}\left[1 - \alpha \cdot \frac{\alpha y}{2\bar{c}}\right].\end{aligned} \tag{21}$$

The assumption $\alpha y/2 \le \bar{c}$ gives $\alpha y/(2\bar{c}) \le 1$, and $\alpha \le 1$, so the bracket is ≥ 0; it is > 0 unless $\alpha = 1$ and $y/2 = \bar{c}$. Hence $\Pi'(q) \ge 0$ and

$$\frac{\partial \beta_\circ}{\partial q} \le 0, \tag{22}$$

strictly except in that knife-edge case, where $L_\circ = y/2$ and $\beta_\circ = K/(y/2)$ does not depend on q. ■ (Interior solution assumed: $\Pi(q) \ge K$, so that $\beta_\circ \le 1$.) The inefficiency $(1-q)\beta_\circ(y - L_\circ)$ in (5.7) is then also decreasing in q. By the product rule, with $L_\circ$ independent of q and $y - L_\circ > 0$ (because $L_\circ \le y/2$),

$$\frac{\partial}{\partial q}\Big[(1-q)\beta_\circ(y - L_\circ)\Big] = (y - L_\circ)\Big[-\beta_\circ + (1-q)\frac{\partial \beta_\circ}{\partial q}\Big] < 0, \tag{23}$$

since $\beta_\circ > 0$ and $\partial\beta_\circ/\partial q \le 0$ by (22).

Answer (sign). $\dfrac{\partial \beta_\circ}{\partial q} = -\dfrac{K\left(\frac{y}{2} - \frac{\alpha^2 y^2}{4\bar{c}}\right)}{\left[q\frac{y}{2} + (1-q)\frac{\alpha^2 y^2}{4\bar{c}}\right]^2} \leq 0$ (strictly negative unless $\alpha = 1$ and $y/2 = \bar{c}$); the incompleteness cost $(1-q)\beta_\circ(y - L_\circ)$ also falls with q.

Intuition. The investor recovers $y/2$ when she threatens a firm that defaulted strategically (it had x and buys itself back) but only $L_\circ \leq y/2$ when the default is a genuine liquidity default (an outside buyer has to be found). A higher q puts more weight on the first, richer recovery, so a smaller liquidation probability suffices to break even — as in the lecture's $1-\gamma_0 = (I-A)/[pE(\tilde{R}_2)+(1-p)L]$ (slide 11).

Common mistakes.

- Letting $L_\circ$ depend on q: it is fixed by the buyer's bargaining and information cost, $L_\circ = \alpha^2 y^2/(4\bar{c})$, so only the weights q and $1-q$ move.
- Signing the derivative without the assumption $\alpha y/2 \leq \bar{c}$: it is what makes $\Pr(c \leq \alpha y/2) = \alpha y/(2\bar{c})$ a probability and gives $L_\circ \leq y/2$.
- Claiming a strict decrease always: at $\alpha = 1$, $y/2 = \bar{c}$ the two recoveries coincide and $\beta_\circ$ does not depend on q.

Book answer. Matches the book's answer (p. 267), which gives (20) with "≤ 0". The book's argument assumes $\alpha y/2 < \bar{c}$ (strict) and writes the step as "$\alpha^2 y/2 < \bar{c}$"; with the text's weak inequality $\alpha y/2 \leq \bar{c}$ the derivative can be zero in the knife-edge case of Step 3.

Ref: Amaro de Matos Ch. 5, §5.2.1 "Extending the Model", pp. 134–137, eq. (5.8); Appendix C.5.2, p. 267.

4. Entrepreneur control and renegotiation — Amaro de Matos 5.1.6 #3, p. 134

What is asked. Under entrepreneur control and a contract with $t_s < \hat{t}$ (so the investor sets $t_s = 0$), show that renegotiating the contract after the state is revealed makes the entrepreneur better off in expectation than sticking to the original contract.

Approach. This is the Aghion–Bolton incomplete-contract model: the contract cannot depend on θ, and with $t_s = 0$ the entrepreneur picks the action with the larger private benefit, which is the wrong action in θ_b. Ex post both parties observe θ, so there is a surplus from switching to a_b that can be shared by a side payment (Coasean renegotiation, as in the lecture's discussion of renegotiation after default). Comparing her payoff with and without the side payment, given that she has all the bargaining power, gives the result.

Notation.

- $q \in (0,1)$: probability of the good state θ_g; a_g, a_b: the actions optimal in θ_g, θ_b.
- $y_j^i \in [0,1]$: expected return in state θ_i under action a_j; l_j^i: entrepreneur's private benefit.
- t_s, $t_s' \geq 0$: variable and fixed parts of the transfer $t(s,r) = t_s r + t_s'$; $\hat{t} = (l_g^b - l_b^b)/(y_b^b - y_g^b)$.
- $\pi_2 = q y_g^g + (1-q) y_g^b$: investor's expected return without renegotiation.
- T: transfer from the investor to the entrepreneur agreed in renegotiation; π_{wt}^E, π_w^E: entrepreneur's expected payoff without and with renegotiation.

Assumption. As in the text, $l_b^b < l_g^b$, $y_b^b > y_g^b$, and a_b is optimal in θ_b: $y_b^b + l_b^b > y_g^b + l_g^b$; in θ_g, $l_g^g > l_b^g$, so the entrepreneur picks a_g there ("the entrepreneur will always choose a_g"). The fixed fee is zero, as π_2 implies (a fixed fee paid in both scenarios would cancel in the comparison). Renegotiation happens after θ is revealed and before the action; the entrepreneur has all the bargaining power and offers to take a_b against a payment T (the text's "willing to pay up to").

Step 1 — Payoff without renegotiation. This is the benchmark. With $t_s = 0$ (and $t'_s = 0$) her reward $t_s y_j^i + t'_s + l_j^i$ reduces to l_j^i, so she takes a_g in both states ($l_g^g > l_b^g$ and $l_g^b > l_b^b$) and gets

$$\pi_{wt}^E = q\, l_g^g + (1-q)\, l_g^b. \tag{24}$$

Step 2 — Room for renegotiation. We find the largest side payment the investor accepts. In θ_g the action is already optimal; nothing to renegotiate. In θ_b switching from a_g to a_b raises the investor's return from y_g^b to y_b^b; paying T leaves her $y_b^b - T$, which beats y_g^b iff $T \le y_b^b - y_g^b$. With all the bargaining power, the entrepreneur asks for the maximum:

$$T = y_b^b - y_g^b. \tag{25}$$

Step 3 — The entrepreneur accepts. She switches to a_b if $T + l_b^b \ge l_g^b$. Substituting (25):

$$\begin{aligned} &(y_b^b - y_g^b) + l_b^b \ge l_g^b \\ \iff\quad &y_b^b + l_b^b \ge y_g^b + l_g^b, \end{aligned} \tag{26}$$

which holds strictly by optimality of a_b in θ_b.

Step 4 — Payoff with renegotiation. In θ_b she now gets $l_b^b + T$. Using (25),

$$\pi_w^E = q\, l_g^g + (1-q)\big[l_b^b + (y_b^b - y_g^b)\big]. \tag{27}$$

Subtracting (24) from (27) (the $q\, l_g^g$ terms cancel):

$$\begin{aligned} \pi_w^E - \pi_{wt}^E &= (1-q)\big[l_b^b + (y_b^b - y_g^b)\big] - (1-q)\, l_g^b \\ &= (1-q)\big[l_b^b + y_b^b - y_g^b - l_g^b\big] \\ &= (1-q)\big[(y_b^b + l_b^b) - (y_g^b + l_g^b)\big] > 0, \end{aligned} \tag{28}$$

by (26) and $q < 1$. The investor's payoff is unchanged:

$$q\, y_g^g + (1-q)\big[y_b^b - (y_b^b - y_g^b)\big] = q\, y_g^g + (1-q)\, y_g^b = \pi_2. \tag{29}$$

So renegotiation raises the entrepreneur's expected payoff by the full efficiency gain in the bad state and leaves the investor at π_2. ■

Answer (gain). $\pi_w^E - \pi_{wt}^E = (1-q)\big[(y_b^b + l_b^b) - (y_g^b + l_g^b)\big] > 0$, with side payment $T = y_b^b - y_g^b$ in θ_b; the investor's expected payoff stays at $\pi_2 = q y_g^g + (1-q) y_g^b$.

Intuition. With $t_s < \hat{t}$ the contract gives the entrepreneur no stake in the return, so in the bad state she picks the action with the larger private benefit, which destroys value. Ex post both parties see the loss and can split the gain by a side payment; with all bargaining power on her side, she captures the whole surplus. Feasibility is unaffected because the investor's payoff stays at π_2.

Common mistakes.

- Giving part of the renegotiation surplus to the investor: the text puts all bargaining power on the entrepreneur, so T is the investor's full willingness to pay and her payoff stays π_2.
- Renegotiating in θ_g too: there a_g is already optimal and there is no surplus.
- Skipping the entrepreneur's acceptance check $T + l_b^b \ge l_g^b$: it holds only because a_b is optimal in θ_b ($y_b^b + l_b^b > y_g^b + l_g^b$).

Book answer. Matches the book's answer (p. 266) in the result. The book's two middle lines carry sign typos: "$= q l_g^g + (1-q)(y_b^b - l_b^b) + (1-q) y_g^b$" should read $q l_g^g + (1-q)(y_b^b + l_b^b) - (1-q) y_g^b$, and "$\ge q l_g^g + (1-q)(y_g^b - l_g^b) + (1-q) y_g^b$" should read $q l_g^g + (1-q)(y_g^b + l_g^b) - (1-q) y_g^b$; the book also writes $\ge$, while the inequality is strict under the text's strict optimality assumption and $q < 1$.

Ref: Amaro de Matos Ch. 5, §5.1.1 "The Model" and §5.1.2 "Entrepreneur Control", pp. 127–131; Appendix C.5.1, p. 266.

5. Diamond (1991): short- vs. long-term debt, no liquidation — Amaro de Matos 5.2.5 #5, p. 148

What is asked. (a) Compute the face value ρ of long-term debt (due at $t = 2$) and r_1 of short-term debt (due at $t = 1$) for the given numbers. (b) Show that no borrower is ever liquidated at $t = 1$. (c) Compute what a good project's entrepreneur gets with long-term debt. (d) Compute the probability e that a good project is downgraded. (e) Compute what a good project's entrepreneur gets with short-term debt (and compare with (c)).

Approach. Diamond's (1991) maturity model: lenders price debt competitively from the public rating, and at $t = 1$ nonverifiable news (an upgrade or a downgrade) reprices short-term debt. Long-term debt is priced once, $\rho q = R^2$; short-term debt by (5.11). Liquidation can only happen if a downgraded borrower cannot roll over, i.e. if $q^d X/R < r_1$ and L beats refinancing, so (b) is a check of that condition. Lemma 5.2.1 then predicts the answer to (c) vs. (e): without liquidation, good types prefer the debt with the higher downgrade value V^d, i.e. short-term debt, because it shifts repayment onto the (mostly bad) downgraded pool.

Notation.

- $L = R = 1$: liquidation value and gross per-period safe return; $\pi = \frac{1}{2}$: success probability of a bad project; $X = \frac{7}{4}$: payoff on success; $C = 0$: control rent.
- $f = \frac{1}{2}$, $f^u = 1$, $f^d = \frac{1}{3}$: prior and posterior probabilities of a good project; e: probability that a good project is downgraded.
- q, q^d: repayment probabilities at $t = 0$ and after a downgrade, (5.9)–(5.10).
- ρ: face value of long-term debt due at $t = 2$; r_1: face value of short-term debt due at $t = 1$; V^d: $t = 1$ value of the debt after a downgrade (V_l^d long, V_s^d short).
- U_l, U_s: expected payoff of a good project under long- and short-term debt.

Step 1 — Repayment probabilities and e. Every later step needs q, q^d and e, so we compute them first. From (5.9) and (5.10),

$$q = \pi + f(1-\pi) = \tfrac{1}{2} + \tfrac{1}{2}\left(\tfrac{1}{2}\right) = \tfrac{3}{4}, \tag{30}$$

$$q^d = \pi + f^d(1-\pi) = \tfrac{1}{2} + \tfrac{1}{3}\left(\tfrac{1}{2}\right) = \tfrac{2}{3}. \tag{31}$$

Bad projects are always downgraded ($f^u = 1$) and good ones with probability e, so a downgrade has probability $fe + (1-f)$ and Bayes' rule gives the downgrade posterior:

$$f^d = \frac{fe}{fe + (1-f)}. \tag{32}$$

Solving (32) for e (multiply out, collect the e terms, divide):

$$\begin{aligned} f^d[fe + (1-f)] &= fe, \\ f^d(1-f) &= fe - f^d fe = fe(1-f^d), \\ e &= \frac{f^d(1-f)}{f(1-f^d)} \\ &= \frac{\left(\frac{1}{3}\right)\left(\frac{1}{2}\right)}{\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)} = \frac{1/6}{1/3} = \tfrac{1}{2}. \end{aligned} \tag{33}$$

Hence the probability of an upgrade is $(1-e)f = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4}$ and of a downgrade $1 - \frac{1}{4} = \frac{3}{4}$.

Step 2 — (a) Face values. Each face value comes from the lenders' break-even condition at its own horizon. *Long-term debt*: the lender lends $K = 1$ for two periods and requires $\rho q = R^2$;

substituting (30),

$$\begin{aligned}\rho\left(\tfrac{3}{4}\right) &= (1)^2,\\ \rho &= \frac{1}{3/4} = \tfrac{4}{3} \approx 1.3333 \leq X = 1.75.\end{aligned} \tag{34}$$

Short-term debt: guess $r_1 \leq \max(L, q^d X/R)$, so that $V_s^d = \min[r_1, \max(L, q^d X/R)] = r_1$ (the debt is repaid in full after a downgrade). Substituting $R = 1$, $(1-e)f = \frac{1}{4}$ and $V_s^d = r_1$ into (5.11):

$$\begin{aligned}r_1 &= \frac{1 - \left[1 - \frac{1}{4}\right](r_1)}{\frac{1}{4}},\\ \tfrac{1}{4}r_1 &= 1 - \tfrac{3}{4}r_1,\\ \tfrac{1}{4}r_1 + \tfrac{3}{4}r_1 &= 1,\\ r_1 &= 1.\end{aligned} \tag{35}$$

Check of the guess: $q^d X/R = \left(\frac{2}{3}\right)\left(\frac{7}{4}\right)/1 = \frac{7}{6} \approx 1.1667$, so $\max(L, q^d X/R) = \frac{7}{6} \geq r_1 = 1$. At $t = 1$ the upgraded borrower rolls r_1 over into debt of face value $r_1 R = 1$, and the downgraded borrower into debt of face value $r_1 R/q^d = 1/\frac{2}{3} = \frac{3}{2}$.

Answer (a). Short-term debt: $r_1 = 1$ (riskless, $r_1 = R$). Long-term debt: $\rho = \frac{4}{3} \approx 1.3333$.

Step 3 — (b) No liquidation. Long-term debt has no payment due at $t = 1$, so there is no default and no liquidation. Short-term debt: after an upgrade the borrower refinances at face value 1; after a downgrade new lenders lend $r_1 = 1$ against face value $\frac{3}{2} \leq X = \frac{7}{4}$, and they break even because $q^d \cdot \frac{3}{2} = \left(\frac{2}{3}\right)\left(\frac{3}{2}\right) = 1 = Rr_1$. Equivalently the refinancing condition $q^d X/R = \frac{7}{6} > r_1 = 1$ holds. Every borrower repays r_1 at $t = 1$, so the liquidation right is never triggered. (Even after a default, lenders would refinance rather than liquidate, since $q^d X/R = \frac{7}{6} > L = 1$.) Both contracts are of the kind covered by Lemma 5.2.1.

Answer (b). No liquidation: long-term debt has no $t = 1$ payment, and with short-term debt every downgraded borrower rolls over $r_1 = 1$ into face value $\frac{3}{2} \leq X = \frac{7}{4}$ (equivalently $q^d X/R = \frac{7}{6} > r_1 = 1$).

Step 4 — (c) Good project, long-term debt. A good project pays X for sure, keeps its control rent C and repays ρ:

$$\begin{aligned}U_l &= X + C - \rho\\ &= \left(\tfrac{7}{4}\right) + (0) - \left(\tfrac{4}{3}\right) = \tfrac{21-16}{12} = \tfrac{5}{12} \approx 0.4167.\end{aligned} \tag{36}$$

Check with the Lemma's formula, using $V_l^d = q^d \rho/R = \left(\frac{2}{3}\right)\left(\frac{4}{3}\right) = \frac{8}{9}$, $R^2/f = 1/\frac{1}{2} = 2$ and $1/f^d - 1/q^d = 3 - \frac{3}{2} = \frac{3}{2}$:

$$\begin{aligned}X + C - \frac{R^2}{f} + RV^d e\left(\frac{1}{f^d} - \frac{1}{q^d}\right) &= \tfrac{7}{4} - 2 + (1)\left(\tfrac{8}{9}\right)\left(\tfrac{1}{2}\right)\left(\tfrac{3}{2}\right)\\ &= -\tfrac{1}{4} + \tfrac{2}{3} = \tfrac{-3+8}{12} = \tfrac{5}{12}.\end{aligned}$$

Answer (c). $U_l = X + C - \rho = \frac{5}{12} \approx 0.4167$.

Step 5 — (d) Probability that a good project is downgraded. From (33), $e = \frac{1}{2}$: half of the good projects are downgraded at $t = 1$ (and all bad ones are), which is what makes $f^d = \frac{1}{3}$. Check with (32): $fe/[fe + (1-f)] = \frac{1}{4}/(\frac{1}{4} + \frac{1}{2}) = \frac{1}{3}$.

Answer (d). $e = \frac{1}{2}$ (the overall downgrade probability is $1 - (1-e)f = \frac{3}{4}$).

Step 6 — (e) Good project, short-term debt. With probability $1-e$ it is upgraded and repays $r_1 R = 1$ at $t=2$; with probability e it is downgraded and repays $RV_s^d/q^d = \frac{3}{2}$ (with $V_s^d = r_1 = 1$). From the Lemma's payoff expression,

$$\begin{aligned} U_s &= e\left(X + C - \frac{RV_s^d}{q^d}\right) + (1-e)(X + C - r_1 R) \\ &= \tfrac{1}{2}\left(\tfrac{7}{4} + 0 - \tfrac{3}{2}\right) + \tfrac{1}{2}\left(\tfrac{7}{4} + 0 - 1\right) \\ &= \tfrac{1}{2}\left(\tfrac{1}{4}\right) + \tfrac{1}{2}\left(\tfrac{3}{4}\right) = \tfrac{1}{2}. \end{aligned} \tag{37}$$

Check: $X+C-R^2/f+RV_s^d e(1/f^d-1/q^d) = -\frac{1}{4}+(1)(1)\left(\frac{1}{2}\right)\left(\frac{3}{2}\right) = -\frac{1}{4}+\frac{3}{4} = \frac{1}{2}$. So $U_s = \frac{1}{2} > U_l = \frac{5}{12}$ (gain $\frac{1}{2} - \frac{5}{12} = \frac{1}{12} \approx 0.0833$): good projects prefer short-term debt, as Lemma 5.2.1 predicts since $V_s^d = 1 > V_l^d = \frac{8}{9}$.

Answer (e). $U_s = \frac{1}{2}$, above $U_l = \frac{5}{12}$ by $\frac{1}{12} \approx 0.0833$: the good entrepreneur prefers short-term debt.

Intuition. Long-term debt locks in the date-0 pooled price $\rho = \frac{4}{3}$, which subsidises the bad projects. Short-term debt lets the price be reset at $t=1$: the upgraded good projects (half of them) refinance at the riskless rate and pay only 1; the downgraded ones pay more ($\frac{3}{2}$), but no project is liquidated, so the reset is pure gain for the good type on average. Short maturity is attractive when refinancing risk is not a liquidation risk.

Common mistakes.

- Pricing long-term debt with R instead of R^2, or with q^d instead of q: at $t=0$ the relevant repayment probability is the prior one, $q = \frac{3}{4}$, over two periods.
- Confusing e (probability that a *good* project is downgraded, $\frac{1}{2}$) with the probability of a downgrade for an unknown type ($\frac{3}{4}$).
- Charging the downgraded good borrower $r_1 R = 1$ instead of $RV_s^d/q^d = \frac{3}{2}$: after a downgrade new lenders price the rollover at the pooled repayment probability q^d.

Book answer. The book gives no worked answer: Appendix C (p. 268) only cites example 1 of Diamond (1991), pp. 716–717, which is not in `Livros/`. The solution above is derived here from §5.2.3 and Lemma 5.2.1. Note that the parameters set $L = 1 = K$, while the text assumes $L < K = 1$; this does not matter here because liquidation never occurs.

Ref: Amaro de Matos Ch. 5, §5.2.3 "The Choice of the Duration", pp. 139–142 (Lemma 5.2.1, p. 142); Appendix C.5.2, p. 268.

6. Optimal contracts in the Bolton–Scharfstein model — Tirole 7.7, p. 325

What is asked. Redo the Bolton–Scharfstein contract with three reward levels instead of one per date-0 outcome. *(PD)*: write the predation-deterrence constraint for this contract. *(IC)*: write the date-0 incentive constraint. *(r_b^S)*: show that the reward after date-0 success without refinancing can be set to zero. *(ordering)*: show that the reward after success–success is at least the reward after failure–success, which is at least $B/\Delta p$.

Approach. Bolton–Scharfstein is the lecture's threat-of-termination logic applied to a firm with a predatory rival: the refinancing probabilities z^S, z^F are the termination threat that disciplines date-0 effort, while predation deterrence requires that threat to be weak. With risk neutrality only *expected* rewards matter, so the proof is a reshuffling argument: move reward to where it relaxes the binding constraints and leaves the others unchanged.

Notation.

- $p_H = 1$, $p_L < 1$, $\Delta p = 1 - p_L$: success probabilities; B_0, B: private benefits from misbehaving at dates 0 and 1 ($B_0 < B$).
- D, M: duopoly and monopoly profits ($M \geq D > 0$); I: investment per date; A: firm 2's net worth.
- z^S, $z^F \in [0,1]$: probabilities of date-1 refinancing after date-0 success (profit D) / failure (profit 0).
- $r_b^S \geq 0$: reward after date-0 success without refinancing; R_b^{SS}, $R_b^{FS} \geq B/\Delta p$: rewards after date-1 success following date-0 success / failure.
- $W^S \equiv z^S R_b^{SS} + (1 - z^S) r_b^S$: entrepreneur's expected reward after date-0 success; $U_b(z^S) = D - I + z^S(D - I)$: NPV.

Step 1 — Date-1 behaviour. Whenever firm 2 is refinanced, $R_b^{SS}, R_b^{FS} \geq B/\Delta p$ make good behaviour optimal at date 1, so firm 2 succeeds with probability $p_H = 1$ and the entrepreneur collects R_b^{SS} or R_b^{FS} for sure.

Step 2 — Generalised (PD). Firm 1's payoffs do not depend on how firm 2 pays its entrepreneur; they depend only on whether firm 2 is active at date 1, which by Step 1 means profit D for firm 1 instead of M. Preying costs firm 1 its date-0 profit D and lowers firm 2's refinancing probability from z^S to z^F, so firm 1's expected date-1 profit rises by $z^F D + (1 - z^F)M - [z^S D + (1 - z^S)M] = (z^S - z^F)(M - D)$. Preying is unprofitable iff the cost D is at least this gain. Hence (PD) is unchanged:

$$D \geq (z^S - z^F)(M - D). \tag{38}$$

Answer (PD). $D \geq (z^S - z^F)(M - D)$, the same as with simple contracts: the rewards $r_b^S, R_b^{SS}, R_b^{FS}$ do not enter it.

Step 3 — Generalised (IC). If the entrepreneur behaves at date 0 (and (PD) holds), date-0 success is certain and she gets W^S. If she misbehaves she gets B_0, succeeds with probability p_L (then W^S), and fails with probability $1 - p_L$ (then $z^F R_b^{FS}$). Behaving is optimal if

$$\begin{aligned} W^S &\geq B_0 + p_L W^S + (1 - p_L) z^F R_b^{FS}, \\ W^S - p_L W^S - (1 - p_L) z^F R_b^{FS} &\geq B_0, \\ (1 - p_L)(W^S - z^F R_b^{FS}) &\geq B_0, \\ (\Delta p)\left[z^S R_b^{SS} + (1 - z^S) r_b^S - z^F R_b^{FS}\right] &\geq B_0, \end{aligned} \tag{39}$$

using $\Delta p = p_H - p_L = 1 - p_L$ and the definition of W^S in the last line. With $r_b^S = 0$, $\mathcal{R}_b^S = z^S R_b^{SS}$ and $\mathcal{R}_b^F = z^F R_b^{FS}$ this is the text's (IC).

Answer (IC). $(\Delta p)\left[z^S R_b^{SS} + (1 - z^S) r_b^S - z^F R_b^{FS}\right] \geq B_0$.

Step 4 — Generalised (IR_l) and the objective. On the equilibrium path date 0 succeeds; investors receive D, pay r_b^S if there is no refinancing, and if there is refinancing invest I, receive D and pay R_b^{SS}:

$$\begin{aligned} & D - (1 - z^S) r_b^S + z^S(D - I - R_b^{SS}) \geq I - A \\ \iff \quad & (D - I) + z^S(D - I) - \left[z^S R_b^{SS} + (1 - z^S) r_b^S\right] + A \geq 0 \\ \iff \quad & U_b(z^S) - W^S + A \geq 0, \end{aligned} \tag{40}$$

where the second line moves $I - A$ to the left and groups terms, and the third uses $U_b(z^S) = D - I + z^S(D - I)$ and the definition of W^S. The entrepreneur's utility is $W^S - A$; with (40) binding it equals $U_b(z^S)$, which is increasing in z^S (Assumption 7.1). The contract maximises $U_b(z^S)$ subject to (38), (39), (40) and the floors $R_b^{SS}, R_b^{FS} \geq B/\Delta p$.

Step 5 — $r_b^S = 0$. The pair (r_b^S, R_b^{SS}) enters (39), (40) and the objective only through W^S, and does not enter (38). Take a contract with $z^S \in (0,1)$ and $r_b^S > 0$, and replace it by

$$\tilde{r}_b^S = 0, \qquad \tilde{R}_b^{SS} = R_b^{SS} + \frac{(1-z^S)r_b^S}{z^S}. \tag{41}$$

Then $z^S\tilde{R}_b^{SS} + (1-z^S)(0) = z^S R_b^{SS} + (1-z^S)r_b^S = W^S$, and $\tilde{R}_b^{SS} > R_b^{SS} \geq B/\Delta p$, so every constraint and the objective are unchanged. If $z^S = 1$, r_b^S is never paid and can be set to 0. Finally $z^S = 0$ is never optimal: from a feasible contract with $z^S = 0$, moving to a small $z^S = \varepsilon > 0$ with $r_b^S = 0$ and $R_b^{SS} = W^S/\varepsilon$ keeps W^S (hence (39)), keeps (38) for $\varepsilon \leq D/(M-D)$, meets the floor $W^S/\varepsilon \geq B/\Delta p$ for small ε (as $W^S \geq B_0/\Delta p > 0$), relaxes (40) because $U_b(\varepsilon) > U_b(0)$, and raises the objective. Hence, without loss of generality, $r_b^S = 0$.

Answer (r_b^S). $r_b^S = 0$ without loss of generality: any $r_b^S > 0$ can be moved into R_b^{SS}, raising it by $(1-z^S)r_b^S/z^S$, with all constraints and the NPV unchanged.

Step 6 — $R_b^{SS} \geq R_b^{FS} \geq B/\Delta p$. R_b^{FS} is paid only off the equilibrium path: it appears in (39) with a negative sign and nowhere else. Lowering it to its floor $B/\Delta p$ relaxes (39) and leaves everything else unchanged, so the optimal contract can set

$$R_b^{FS} = \frac{B}{\Delta p} \leq R_b^{SS}. \tag{42}$$

The book adds that (39) binds when (38) binds (otherwise z^F could be raised to relax (PD)); then $R_b^{FS} > B/\Delta p$ would leave (39) slack, so (42) holds at every optimum. With (PD) binding, $z^F = z^S - D/(M-D)$ and $r_b^S = 0$, substituting into (39):

$$\begin{aligned} z^S R_b^{SS} - \left(z^S - \frac{D}{M-D}\right) R_b^{FS} &= \frac{B_0}{\Delta p}, \\ z^S(R_b^{SS} - R_b^{FS}) + \frac{D}{M-D} R_b^{FS} &= \frac{B_0}{\Delta p}, \end{aligned} \tag{43}$$

where the first line divides (39) (with equality) by Δp and the second expands $-(z^S - \frac{D}{M-D})R_b^{FS}$ and groups the z^S terms.

Answer (ordering). $R_b^{FS} = B/\Delta p$ (its floor) and $R_b^{SS} \geq B/\Delta p = R_b^{FS}$, so $R_b^{SS} \geq R_b^{FS} \geq B/\Delta p$.

Intuition. Under risk neutrality only expected rewards matter, so the entrepreneur should be paid where it helps incentives most — in the continuation branch after success, which also ties her pay to the date-1 output. Paying her after date-0 failure only weakens date-0 incentives; the contract does it only because predation deterrence forces refinancing after failure ($z^F > 0$), and then the date-1 incentive floor $B/\Delta p$ is the least it can pay.

Common mistakes.

- Setting $R_b^{FS} = 0$: once the firm is refinanced after failure, the date-1 incentive constraint still requires $R_b^{FS} \geq B/\Delta p$; the floor, not zero, is the minimum.
- Putting the rewards into (PD): firm 1 cares only about whether firm 2 is active at date 1, so (PD) depends on $z^S - z^F$ alone.
- Writing the IC with a generic p_H: here $p_H = 1$, so the date-0 success probability gap is $\Delta p = 1 - p_L$ and success is certain on the equilibrium path.

Book answer. Matches the book's answer (pp. 603–604): same (PD), (IC′) and (IR′), $r_b^S = 0$ by shifting reward into R_b^{SS}, and $R_b^{FS} = B/\Delta p$. The book argues $R_b^{SS} \geq R_b^{FS}$ via the binding (IC′) and leaves aside $z^S = 0$; Step 5 covers that case and Step 6 gives the direct argument.

Ref: Tirole Ch. 7, §7.1.2.1 "Financial Muscle and Predation" (Bolton–Scharfstein), pp. 290–294; Answers, pp. 603–604.

7. Liquidation incentives with a private signal — Tirole 3.14, pp. 149–150

What is asked. (i) With a verifiable interim signal γ: explain why the entrepreneur's pay should depend only on γ, compute the pledgeable income, and show that the project is financed iff $A \geq \bar{A}$. (ii) With γ seen only by the entrepreneur: design the cheapest scheme (R_b, L_b) that makes her both behave and report γ truthfully, and show that financing now requires $A \geq \bar{A} + \underline{\gamma}B/[(\Delta p)(\Delta\gamma)]$.

Approach. Fixed-investment model of Tirole §3.2 (pledgeable income, $\mathcal{P} \geq I - A$) with an interim liquidation option. In (i) the signal is a sufficient statistic for effort, so pay on γ and liquidate after bad news. In (ii) the liquidation decision rests on information only the insider has — the Topic 5 problem of nonverifiable information — so the contract must add truth-telling constraints: the entrepreneur must not gain by hiding bad news to avoid termination. The extra rent this requires is the increase in the net-worth threshold.

Notation.

- $A < I$: entrepreneur's cash and the investment; R: final profit on success; L: liquidation value, fully pledgeable.
- $\gamma \in \{\underline{\gamma}, \bar{\gamma}\}$: success probability revealed by the signal; $\Delta\gamma = \bar{\gamma} - \underline{\gamma} > 0$; $\bar{\gamma}R > L > \underline{\gamma}R$.
- p_H, p_L, $\Delta p = p_H - p_L > 0$: probability of $\gamma = \bar{\gamma}$ if she behaves / misbehaves; B: private benefit.
- (i) $w(\gamma) \geq 0$: reward after signal γ; $\mathcal{P}$: pledgeable income; $\bar{A}$: financing threshold.
- (ii) R_b: reward if she announces $\bar{\gamma}$ and the profit is R; L_b: reward if she announces $\underline{\gamma}$ (liquidation); $\mathcal{P}'$: pledgeable income.

Step 1 — (i) Rewards depend on γ only. Effort affects only the distribution of γ; once γ is known, the final outcome is pure noise for the entrepreneur (γ is a sufficient statistic for effort). With risk neutrality, any scheme that pays according to γ and the final profit acts on effort and on the investors' cost only through the expected reward given γ, $w(\gamma)$. Paying $w(\gamma)$ directly leaves both unchanged, so rewards can be made contingent on γ alone.

Step 2 — (i) Liquidation policy. With γ verifiable, the contract liquidates if and only if $\gamma = \underline{\gamma}$: this raises investors' income by $L - \underline{\gamma}R > 0$ after a bad signal, and continuing after a good one gives $\bar{\gamma}R > L$. The rewards are transfers that do not depend on this choice, so it maximises both NPV and pledgeable income.

Step 3 — (i) Incentive constraint and pledgeable income. Pledgeable income is total expected income minus the smallest expected rent compatible with effort, so we first find that rent. Behaving is optimal if

$$\begin{aligned} p_H w(\bar{\gamma}) + (1-p_H)w(\underline{\gamma}) &\geq p_L w(\bar{\gamma}) + (1-p_L)w(\underline{\gamma}) + B, \\ (p_H - p_L)w(\bar{\gamma}) - \big[(1-p_L) - (1-p_H)\big]w(\underline{\gamma}) &\geq B, \\ (\Delta p)\big[w(\bar{\gamma}) - w(\underline{\gamma})\big] &\geq B. \end{aligned} \tag{44}$$

The expected rent $p_H w(\bar{\gamma}) + (1-p_H)w(\underline{\gamma})$ rises in both rewards, so the cheapest scheme sets $w(\underline{\gamma}) = 0$ (limited liability) and $w(\bar{\gamma}) = B/\Delta p$, an expected rent $p_H B/\Delta p$. With the liquidation policy of Step 2, total expected income is $p_H\bar{\gamma}R + (1-p_H)L$. Hence

$$\mathcal{P} = p_H\bar{\gamma}R + (1-p_H)L - p_H\frac{B}{\Delta p}. \tag{45}$$

Step 4 — (i) Financing condition. Investors lend if $\mathcal{P} \geq I - A$. Using (45) and the definition

of $\bar{A}$, $p_H(\bar{\gamma}R - B/\Delta p) + (1-p_H)L = I - \bar{A}$:

$$\begin{aligned} p_H\left(\bar{\gamma}R - \frac{B}{\Delta p}\right) + (1-p_H)L \geq I - A \\ \iff \quad I - \bar{A} \geq I - A \\ \iff \quad A \geq \bar{A}. \end{aligned} \tag{46}$$

Answer (i). Pay on γ only, $w(\bar{\gamma}) = B/\Delta p$, $w(\underline{\gamma}) = 0$, liquidate iff $\gamma = \underline{\gamma}$; $\mathcal{P} = p_H(\bar{\gamma}R - B/\Delta p) + (1-p_H)L$; financed iff $A \geq \bar{A}$.

Step 5 — (ii) Truth-telling constraints. Now only the entrepreneur sees γ. Announcing $\bar{\gamma}$ gives continuation and an expected reward γR_b; announcing $\underline{\gamma}$ gives liquidation and L_b. Truth-telling in each state requires

$$\bar{\gamma}R_b \geq L_b, \tag{47}$$

$$L_b \geq \underline{\gamma}R_b. \tag{48}$$

Step 6 — (ii) Effort constraint. Given truthful reports, she gets $\bar{\gamma}R_b$ after a good signal and L_b after a bad one, so, as in (44),

$$(\Delta p)(\bar{\gamma}R_b - L_b) \geq B. \tag{49}$$

Step 7 — (ii) Cheapest scheme. Investors receive $p_H\bar{\gamma}(R - R_b) + (1-p_H)(L - L_b)$, so they minimise the expected rent $p_H\bar{\gamma}R_b + (1-p_H)L_b$. Both R_b and L_b should be as low as (48) and (49) allow; making both bind,

$$L_b = \underline{\gamma}R_b, \tag{50}$$

$$\begin{aligned} (\Delta p)(\bar{\gamma}R_b - (\underline{\gamma}R_b)) &= B, \\ R_b &= \frac{B}{(\Delta p)(\Delta\gamma)}, \\ L_b &= \underline{\gamma}\,\frac{B}{(\Delta p)(\Delta\gamma)}. \end{aligned} \tag{51}$$

(Formally: with $\delta \equiv \bar{\gamma}R_b - L_b \geq B/\Delta p$, (48) is $L_b \geq \underline{\gamma}\delta/\Delta\gamma$, so the rent $L_b + p_H\delta \geq (\underline{\gamma}/\Delta\gamma + p_H)\delta$ is minimised at $\delta = B/\Delta p$.) (47) holds since $\bar{\gamma}R_b - L_b = B/\Delta p > 0$.

Step 8 — (ii) Pledgeable income and financing condition. Substituting (50)–(51) into the rent, and writing $\bar{\gamma} = \underline{\gamma} + \Delta\gamma$ in the second line,

$$\begin{aligned} p_H\bar{\gamma}R_b + (1-p_H)L_b &= \left[p_H\bar{\gamma} + (1-p_H)\underline{\gamma}\right]\frac{B}{(\Delta p)(\Delta\gamma)} \\ &= \left[p_H\underline{\gamma} + p_H\Delta\gamma + \underline{\gamma} - p_H\underline{\gamma}\right]\frac{B}{(\Delta p)(\Delta\gamma)} \\ &= \left[\underline{\gamma} + p_H\Delta\gamma\right]\frac{B}{(\Delta p)(\Delta\gamma)} \\ &= p_H\frac{B}{\Delta p} + \underline{\gamma}\,\frac{B}{(\Delta p)(\Delta\gamma)}. \end{aligned} \tag{52}$$

So, subtracting this rent from total expected income and using (45),

$$\begin{aligned} \mathcal{P}' &= p_H\bar{\gamma}R + (1-p_H)L - p_H\frac{B}{\Delta p} - \underline{\gamma}\,\frac{B}{(\Delta p)(\Delta\gamma)} \\ &= \mathcal{P} - \underline{\gamma}\,\frac{B}{(\Delta p)(\Delta\gamma)}, \end{aligned} \tag{53}$$

and $\mathcal{P}' \geq I - A$ is, using $\mathcal{P} = I - \bar{A}$ from (46),

$$I - \bar{A} - \underline{\gamma}\,\frac{B}{(\Delta p)(\Delta\gamma)} \geq I - A$$
$$\iff \quad A \geq \bar{A} + \underline{\gamma}\,\frac{B}{(\Delta p)(\Delta\gamma)}. \qquad \blacksquare \tag{54}$$

(Continuing after both signals would avoid the truth-telling problem but costs the same rent, since then $(\Delta p)(\Delta\gamma)R_b \geq B$, and loses $(1-p_H)(L-\underline{\gamma}R)$ of pledgeable income; so truthful liquidation is better. Implicitly $R_b \leq R$ and $L_b \leq L$.)

Answer (ii). $R_b = \dfrac{B}{(\Delta p)(\Delta\gamma)}$, $L_b = \underline{\gamma}R_b$; pledgeable income falls by $\underline{\gamma}B/[(\Delta p)(\Delta\gamma)]$, so the project is funded iff $A \geq \bar{A} + \underline{\gamma}\,\dfrac{B}{(\Delta p)(\Delta\gamma)}$.

Intuition. When the signal is private, the entrepreneur must be paid to admit bad news, because admitting it kills the project in which she has a stake: she gets $L_b = \underline{\gamma}R_b$, a "golden parachute" worth what she would expect by hiding the news. That raises the stake needed for effort from $B/\Delta p$ to $B/[(\Delta p)(\Delta\gamma)]$ and the expected rent by $\underline{\gamma}B/[(\Delta p)(\Delta\gamma)]$; the net-worth threshold rises one for one. It vanishes when $\underline{\gamma} = 0$ (nothing to hide) and grows as the signal becomes less informative ($\Delta\gamma \to 0$).

Common mistakes.

- Setting $L_b = 0$ in (ii): then an entrepreneur with bad news reports $\bar{\gamma}$ (she gets $\underline{\gamma}R_b > 0$ by continuing), the project is never liquidated and the L income is lost.
- Writing the effort constraint as $(\Delta p)\bar{\gamma}R_b \geq B$: after a bad signal she still gets L_b, so what effort buys is the *difference* $\bar{\gamma}R_b - L_b$.
- Counting L as pledgeable after a good signal or $\bar{\gamma}R$ after a bad one: the efficient policy continues iff $\gamma = \bar{\gamma}$ because $\bar{\gamma}R > L > \underline{\gamma}R$.

Book answer. Matches the book's answer (pp. 577–578): $L_b = \underline{\gamma}R_b$, $(\Delta p)(\bar{\gamma}R_b - L_b) = B$, and the threshold $\bar{A} + \underline{\gamma}B/[(\Delta p)(\Delta\gamma)]$.

Ref: Tirole Ch. 3, §3.2 "The Fixed-Investment Model" and §3.2.2 "The Lenders' Credit Analysis", pp. 115–117; Exercise 3.14, pp. 149–150; Answers, pp. 577–578.

8. Decreasing returns to scale and debt maturity — Tirole Exercise 5.7, p. 232

What is asked. (i) In the liquidity model with short-term income rI and a continuum of reinvestment shocks ρI, but with a concave success payoff $R(I)$, derive the first-order conditions that pin down the scale I and the continuation cutoff ρ^*. (ii) Under free cash flow ($r > \rho^*$), find how a stronger balance sheet (higher A) changes debt maturity, i.e. the short-term debt the firm must pay at date 1 per unit of investment.

Approach. Holmström–Tirole liquidity model (Tirole §5.3.2 and §5.6.1): the borrower maximises the NPV subject to the lenders' break-even, a Lagrangian problem in (I, ρ^*). Continuing after a shock ρ costs ρI and is worth ρ_1 in value but only ρ_0 in pledgeable income, so the optimal cutoff lies between them. Under free cash flow the cutoff is implemented by short-term debt that pumps $(r-\rho^*)I$ out of the firm — the termination threat of the lecture, here a liquidity sweep — so maturity moves with ρ^*, and comparative statics in A answer (ii).

Notation.

- $I > 0$: investment (endogenous); $A \geq 0$: the entrepreneur's cash; lenders are competitive, rate of return 0.
- $R(I)$: date-2 income in case of continuation and success, $R(0) = 0$, $R' > 0$, $R'' < 0$.

- rI: short-term (date-1) income, verifiable and fully pledgeable, $r > 0$.
- $\rho \geq 0$: per-unit liquidity shock (reinvestment need ρI), c.d.f. F, density f on $[0, \infty)$; ρ^*: continuation cutoff (continue iff $\rho \leq \rho^*$).
- $p_H > p_L$: success probabilities when behaving / misbehaving, $\Delta p = p_H - p_L$; BI: private benefit from misbehaving; R_b: entrepreneur's stake in success.
- $\rho_1(I) \equiv p_H R(I)/I$ and $\rho_0(I) \equiv p_H[R(I)/I - B/\Delta p]$: per-unit expected income and per-unit pledgeable income at scale I; $\rho_1 - \rho_0 = p_H B/\Delta p$.
- $\mu \geq 0$: Lagrange multiplier of the lenders' breakeven constraint.

Step 1 — The program. We set up the borrower's problem. As in Section 5.6.1, the incentive constraint $(\Delta p)R_b \geq BI$ binds (any larger stake only lowers pledgeable income), so the income pledgeable in case of success is $R(I) - BI/\Delta p$. With competitive lenders the entrepreneur gets the NPV: short-term income, plus expected date-2 income when continuing (probability $F(\rho^*)$), minus the investment and the expected reinvestment,

$$U_b = rI + F(\rho^*)\, p_H R(I) - \Big[1 + \int_0^{\rho^*} \rho f(\rho)\, d\rho\Big] I, \tag{55}$$

subject to the lenders' breakeven constraint

$$rI + F(\rho^*)\, p_H \Big[R(I) - \frac{BI}{\Delta p}\Big] \geq I - A + \Big[\int_0^{\rho^*} \rho f(\rho)\, d\rho\Big] I. \tag{56}$$

The Lagrangian is $\mathcal{L} = U_b + \mu\,[\text{LHS} - \text{RHS of (56)}]$.

Step 2 — (i) First-order conditions. We differentiate $\mathcal{L}$ with respect to each choice variable. By the fundamental theorem of calculus, $\frac{d}{d\rho^*}\int_0^{\rho^*} \rho f\, d\rho = \rho^* f(\rho^*)$, and $\frac{d}{d\rho^*}F(\rho^*) = f(\rho^*)$. Term by term,

$$\begin{aligned} \frac{\partial U_b}{\partial \rho^*} &= f(\rho^*)\, p_H R(I) - \rho^* f(\rho^*)\, I, \\ \frac{\partial[\text{LHS} - \text{RHS of (56)}]}{\partial \rho^*} &= f(\rho^*)\, p_H \Big[R(I) - \frac{BI}{\Delta p}\Big] - \rho^* f(\rho^*)\, I, \end{aligned}$$

so $\partial\mathcal{L}/\partial\rho^* = 0$ reads

$$0 = f(\rho^*)\big[p_H R(I) - \rho^* I\big] + \mu f(\rho^*)\Big[p_H R(I) - \frac{p_H BI}{\Delta p} - \rho^* I\Big].$$

Dividing by $f(\rho^*)I > 0$ and using $\rho_1(I) = p_H R(I)/I$ and $\rho_0(I) = p_H[R(I)/I - B/\Delta p]$,

$$0 = \big[\rho_1(I) - \rho^*\big] + \mu\big[\rho_0(I) - \rho^*\big]. \tag{57}$$

Collecting the ρ^* terms, $(1+\mu)\rho^* = \rho_1(I) + \mu\rho_0(I)$, so $\rho^* = \frac{\rho_1(I) + \mu\rho_0(I)}{1+\mu}$, a weighted average that lies between $\rho_0(I)$ and $\rho_1(I)$, the analogue of (5.8). With respect to I, the derivatives are

$$\begin{aligned} \frac{\partial U_b}{\partial I} &= r + F(\rho^*) p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho, \\ \frac{\partial[\text{LHS} - \text{RHS of (56)}]}{\partial I} &= r + F(\rho^*) p_H \Big(R'(I) - \frac{B}{\Delta p}\Big) - 1 - \int_0^{\rho^*} \rho f\, d\rho, \end{aligned}$$

so $\partial\mathcal{L}/\partial I = 0$ is

$$0 = \Big[r + F(\rho^*) p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho\Big] + \mu\Big[r + F(\rho^*) p_H \Big(R'(I) - \frac{B}{\Delta p}\Big) - 1 - \int_0^{\rho^*} \rho f\, d\rho\Big]. \tag{58}$$

These two conditions, together with (56) holding with equality, determine (I, ρ^*, μ).

Step 3 — (i) Eliminating μ. The multiplier is not observable, so we combine the two conditions into one that involves only (I, ρ^*). Write the second bracket of (58) as the first minus $F(\rho^*)p_H B/\Delta p$ and collect:

$$(1+\mu)\Big[r + F(\rho^*)p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho\Big] = \mu F(\rho^*)\frac{p_H B}{\Delta p}. \tag{59}$$

From (57), $\mu(\rho^* - \rho_0) = \rho_1 - \rho^*$, so $\mu = \frac{\rho_1 - \rho^*}{\rho^* - \rho_0}$ and $1 + \mu = \frac{\rho^* - \rho_0 + \rho_1 - \rho^*}{\rho^* - \rho_0} = \frac{\rho_1 - \rho_0}{\rho^* - \rho_0}$, hence $\frac{\mu}{1+\mu} = \frac{\rho_1 - \rho^*}{\rho_1 - \rho_0}$. Since $\rho_1 - \rho_0 = p_H \frac{R(I)}{I} - p_H\Big(\frac{R(I)}{I} - \frac{B}{\Delta p}\Big) = \frac{p_H B}{\Delta p}$,

$$\frac{\mu}{1+\mu}\,\frac{p_H B}{\Delta p} = \frac{\rho_1(I) - \rho^*}{\rho_1(I) - \rho_0(I)}\,(\rho_1(I) - \rho_0(I)) = \rho_1(I) - \rho^*. \tag{60}$$

Dividing (59) by $1 + \mu$ and substituting (60):

$$r + F(\rho^*)p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho = F(\rho^*)\Big(\frac{p_H R(I)}{I} - \rho^*\Big).$$

Move the R terms to the left and the rest to the right:

$$\begin{aligned} F(\rho^*)\, p_H\Big[R'(I) - \frac{R(I)}{I}\Big] &= 1 - r + \int_0^{\rho^*} \rho f\, d\rho - \rho^* F(\rho^*) \\ &= 1 - r - \Big[\rho^* F(\rho^*) - \int_0^{\rho^*} \rho f\, d\rho\Big]. \end{aligned}$$

The bracket is $\int_0^{\rho^*}(\rho^* - \rho) f(\rho)\, d\rho$; integrating by parts with $F(0) = 0$, $\int_0^{\rho^*}(\rho^* - \rho) f\, d\rho = \big[(\rho^* - \rho)F(\rho)\big]_0^{\rho^*} + \int_0^{\rho^*} F(\rho)\, d\rho = \int_0^{\rho^*} F(\rho)\, d\rho$. Dividing by $F(\rho^*) > 0$,

$$p_H\Big[R'(I) - \frac{R(I)}{I}\Big] = \frac{1 - r - \int_0^{\rho^*} F(\rho)\, d\rho}{F(\rho^*)}. \tag{61}$$

The binding (56), solved for A and with $p_H[R(I) - BI/\Delta p] = I\rho_0(I)$, reads

$$\begin{aligned} A &= I - rI + \Big[\int_0^{\rho^*} \rho f(\rho)\, d\rho\Big] I - F(\rho^*)\, I\rho_0(I) \\ &= I\Big[1 - r + \int_0^{\rho^*} \rho f(\rho)\, d\rho - F(\rho^*)\rho_0(I)\Big]. \end{aligned} \tag{62}$$

Check: with constant returns, $R(I) = RI$, the left side of (61) is 0 and (61) becomes $\int_0^{\rho^*} F = 1 - r$, which is (5.6″). With strict concavity and $R(0) = 0$, $R(I)/I > R'(I)$, so the left side is negative and $\int_0^{\rho^*} F > 1 - r$: the marginal unit of scale is worth less than the average one, so at the margin the firm spends more on continuation and less on scale than (5.6″) would say.

Answer (i). The FOCs are (57), $\rho_1(I) - \rho^* + \mu[\rho_0(I) - \rho^*] = 0$, and (58); eliminating μ, the optimum (I, ρ^*) solves $p_H[R'(I) - R(I)/I] = [1 - r - \int_0^{\rho^*} F(\rho)\, d\rho]/F(\rho^*)$ together with the binding break-even (62), with $\rho_0(I) < \rho^* < \rho_1(I)$.

Step 4 — (ii) I and ρ^* move together. Condition (61) links the two choices; we find the sign of that link. Write $\varphi(I) \equiv p_H[R(I)/I - R'(I)] > 0$, increasing by assumption, and $h(\rho^*)$ for the right side of (61), so (61) is $-\varphi(I) = h(\rho^*)$. The numerator of h is $N(\rho^*) = 1 - r - \int_0^{\rho^*} F$, with $N'(\rho^*) = -F(\rho^*)$; by the quotient rule,

$$\begin{aligned} h'(\rho^*) &= \frac{N'(\rho^*)F(\rho^*) - N(\rho^*)f(\rho^*)}{F(\rho^*)^2} \\ &= \frac{-F(\rho^*)^2 - f(\rho^*)\big[1 - r - \int_0^{\rho^*} F\big]}{F(\rho^*)^2} \\ &= -1 - \frac{f(\rho^*)}{F(\rho^*)}\, h(\rho^*). \end{aligned} \tag{63}$$

Assumption. Tirole's answer takes h decreasing. Since $h<0$ at the optimum, (63) is negative iff $f(\rho^*)\big[\int_0^{\rho^*} F-(1-r)\big]<F(\rho^*)^2$. This holds, e.g., for F uniform on $[0,\bar\rho]$ ($f=1/\bar\rho$, $F(\rho^*)=\rho^*/\bar\rho$, $\int_0^{\rho^*}F=\rho^{*2}/(2\bar\rho)$): multiplying by $\bar\rho^2$, the condition reads $\rho^{*2}/2-\bar\rho(1-r)<\rho^{*2}$, always true for $r\le 1$. We assume it.

Then along (61) an increase in I lowers $-\varphi(I)$, which requires a lower h, i.e. a higher ρ^*. Totally differentiating $-\varphi(I)=h(\rho^*)$ gives $-\varphi'(I)\,dI=h'(\rho^*)\,d\rho^*$, so (61) defines $\rho^*=g(I)$ with

$$g'(I)=\frac{-\varphi'(I)}{h'(\rho^*)}>0. \tag{64}$$

Step 5 — (ii) Effect of A. The binding break-even (62) tells how much net worth a given (I,ρ^*) requires; we differentiate its right side. With respect to ρ^*,

$$\begin{aligned}\frac{\partial A}{\partial \rho^*}&=I\big[\rho^* f(\rho^*)-f(\rho^*)\rho_0(I)\big]\\&=If(\rho^*)\big[\rho^*-\rho_0(I)\big]>0,\end{aligned} \tag{65}$$

positive because $\rho^*>\rho_0(I)$ by (57). With respect to I (using $\frac{d}{dI}[I\rho_0(I)]=p_H[R'(I)-B/\Delta p]$),

$$\frac{\partial A}{\partial I}=1-r+\int_0^{\rho^*}\rho f\,d\rho-F(\rho^*)p_H\Big[R'(I)-\frac{B}{\Delta p}\Big].$$

The second bracket of (58) is $-\partial A/\partial I$ and the first is $-\partial A/\partial I+F(\rho^*)p_HB/\Delta p$, so (58) reads $-(1+\mu)\,\partial A/\partial I+F(\rho^*)p_HB/\Delta p=0$, i.e.

$$\frac{\partial A}{\partial I}=\frac{F(\rho^*)\,p_HB}{(1+\mu)\Delta p}>0. \tag{66}$$

Along the optimum, $\rho^*=g(I)$, so by (64), (65) and (66)

$$\frac{dA}{dI}=\frac{\partial A}{\partial I}+\frac{\partial A}{\partial \rho^*}\,g'(I)>0, \tag{67}$$

hence $dI/dA=1/(dA/dI)>0$ and $d\rho^*/dA=g'(I)\,dI/dA>0$: a weaker balance sheet lowers both the scale and the cutoff.

Step 6 — (ii) Debt maturity. Under the free-cash-flow assumption $r>\rho^*$ the optimum is implemented as in Section 5.6.1(a): the firm keeps ρ^*I to meet shocks $\rho\le\rho^*$, must pay out $P_1=(r-\rho^*)I$ at date 1 (short-term debt), is barred from diluting the initial investors, and the investors' long-term claim is $R(I)-BI/\Delta p$ in case of success. Per unit of investment the short-term debt is $r-\rho^*$, and by Step 5

$$\frac{d(r-\rho^*)}{dA}=-\frac{d\rho^*}{dA}<0. \tag{68}$$

A firm with a weaker balance sheet (lower A) issues more short-term debt per unit of investment: its debt maturity is shorter. (The level $(r-\rho^*)I$ can move either way, because I falls too.)

Answer (ii). $dI/dA>0$ and $d\rho^*/dA>0$, so the short-term debt per unit of investment, $r-\rho^*$, falls with A: weaker balance sheets carry shorter maturity (more of the date-1 cash flow is paid out), provided $h'(\rho^*)<0$ (true, e.g., for uniform shocks).

Intuition. A weak balance sheet makes pledgeable income scarce. The firm raises it by continuing less often, i.e. by keeping less liquidity in the firm, which means sweeping more of the date-1 cash flow out to creditors through short-term debt. Decreasing returns add one more channel: a lower scale raises $R(I)/I$ and hence per-unit pledgeable income, and the two margins (scale and continuation) are cut together.

Common mistakes.

- Carrying over the constant-returns result $\int_0^{\rho^*} F = 1 - r$, under which ρ^* does not depend on A: with $R'' < 0$ the left side of (61) is negative and moves with I, which is what makes ρ^* depend on A.
- Stating the FOCs without the multiplier on the break-even: the cutoff is a weighted average of ρ_1 (value) and ρ_0 (pledgeable income), not ρ_1 alone.
- Reading "shorter maturity" off the level $(r - \rho^*)I$: since I also falls with A, the unambiguous statement is per unit of investment.

Book answer. Matches the book's answer (p. 593): same condition (61) (the book writes its numerator as $1 - r - \int_0^{\rho^*}(\rho^* - \rho)f(\rho)d\rho$, equal to ours by integration by parts) and the same conclusion (lower $A \Rightarrow$ lower I and $\rho^* \Rightarrow$ more short-term debt). The book states without proof that the right side of (61) is decreasing in ρ^*; (63) shows this needs a condition on F (true for the uniform).

Ref: Tirole §5.3.2 "Continuum of Liquidity Shocks", pp. 209–210; §5.6.1 "Optimal Claim Maturity", p. 226; Exercise 5.7, p. 232; answer p. 593.

9. The priority game: uncoordinated lending and short-term bias — Tirole Exercise 5.9, p. 233

What is asked. (i) Find the first-best date-1 deepening investment I_1^* and show that a mix of short- and long-term debt implements it, with some indeterminacy in how the claims (and the entrepreneur's pay) are split. (ii) With an opaque balance sheet and many lenders, show that the entrepreneur gains by secretly swapping long-term claim for short-term claim with one lender, and say whether this pins down the split left open in (i).

Approach. Without moral hazard the borrower captures the NPV, so (i) is a plain optimisation plus counting equations against unknowns. In (ii) the date-1 payout to one lender is paid out of the reinvestment fund, lowering the success probability of *every* long-term claim: an externality that neither party to the side deal internalises (a common-agency, "priority" problem). Only collusion-proof structures survive, which is the lecture's point that maturity is chosen to discipline what the insider can do with unverifiable cash.

Notation.

- I: fixed date-0 investment; $A = 0$; no moral hazard (private benefit 0); risk-free rate 0.
- $r > 0$: known date-1 return; R or 0: date-2 return, success probability $p + \tau(I_1)$.
- $I_1 \geq 0$: date-1 deepening investment; τ increasing, concave, $\tau'(0) = \infty$, $\tau'(r)R < 1$.
- $r_l, r_b \geq 0$: date-1 payments to lenders (short-term debt) and to the entrepreneur; $R_l, R_b \geq 0$: success-contingent date-2 payments, $R_l + R_b = R$.
- I_1^*: first-best deepening investment; $\delta r_l, \delta R_l > 0$: sizes of a secret deal.

Step 1 — (i) First best. The first best maximises total surplus over I_1. The NPV is date-1 cash not reinvested plus expected date-2 income minus the initial investment:

$$\text{NPV}(I_1) = r - I_1 + [p + \tau(I_1)]R - I. \tag{69}$$

Its derivatives are $\text{NPV}'(I_1) = -1 + \tau'(I_1)R$ and $\text{NPV}''(I_1) = \tau''(I_1)R \leq 0$, so it is concave and the first-order condition $\text{NPV}'(I_1^*) = 0$ is sufficient:

$$-1 + \tau'(I_1^*)\,R = 0 \quad \Longleftrightarrow \quad \tau'(I_1^*)\,R = 1. \tag{70}$$

$\tau'(0) = \infty$ gives $I_1^* > 0$; $\tau'(r)R < 1 = \tau'(I_1^*)R$ and τ' decreasing give $I_1^* < r$, so $r - I_1^* > 0$ can be paid out at date 1.

Step 2 — (i) Implementation and indeterminacy. With transparent books the contract fixes the date-1 payouts, and hence $I_1 = r - r_l - r_b$. The first best requires

$$r_b + r_l + I_1^* = r, \tag{71}$$
$$R_b + R_l = R, \tag{72}$$
$$I = r_l + [p + \tau(I_1^*)]R_l, \tag{73}$$

the last being the lenders' breakeven condition. Four unknowns (r_b, r_l, R_b, R_l) and three equations: one degree of freedom. For instance (when $r - I_1^* < I$), maximal short-term debt $r_b = 0$, $r_l = r - I_1^*$ with $R_l = \frac{I-(r-I_1^*)}{p+\tau(I_1^*)}$; or pure long-term debt $r_l = 0$, $R_l = \frac{I}{p+\tau(I_1^*)}$ (if this is $\leq R$), with the entrepreneur taking $r_b = r - I_1^*$. Without moral hazard the entrepreneur gets the first-best NPV in all cases, so the split of her pay between r_b and R_b, and of the lenders' claims between short and long term, is undetermined.

Answer (i). I_1^* solves $\tau'(I_1^*)R = 1$, with $0 < I_1^* < r$; any $(r_b, r_l, R_b, R_l) \geq 0$ with $r_b + r_l = r - I_1^*$, $R_b + R_l = R$ and $r_l + [p+\tau(I_1^*)]R_l = I$ implements it — one degree of freedom in the short-/long-term split.

Step 3 — (ii) The secret deal. With $r - I_1^* < I$, (73) and $r_l \leq r - I_1^*$ force $R_l > 0$, hence $R_b < R$ by (72). Starting from any first-best structure, the entrepreneur offers one lender $+\delta r_l$ in short-term claim against the transfer to herself of δR_l of his long-term claim. The lender is atomistic, takes I_1 as given, and accepts iff

$$\delta r_l \geq [p + \tau(I_1)]\, \delta R_l. \tag{74}$$

Date-1 cash is fixed and the other claims cannot be defrauded, so the extra payout comes out of deepening investment:

$$\delta I_1 = -\delta r_l. \tag{75}$$

The entrepreneur's utility is $U_b = r_b + [p + \tau(I_1)]R_b$; in the deal r_b is unchanged and she receives the long-term claim the lender gives up, $\delta R_b = \delta R_l$. To first order (product rule),

$$\delta U_b = \tau'(I_1)\,(\delta I_1)\, R_b + [p + \tau(I_1)]\,(\delta R_l).$$

Substituting (75) and (74) at equality ($\delta R_l = \delta r_l/[p + \tau(I_1)]$),

$$\begin{aligned} \delta U_b &= \tau'(I_1)\,(-\delta r_l)\, R_b + [p + \tau(I_1)]\Big(\frac{\delta r_l}{p + \tau(I_1)}\Big) \\ &= \big[1 - \tau'(I_1)R_b\big]\, \delta r_l. \end{aligned} \tag{76}$$

At $I_1 = I_1^*$, (70) gives $\tau'(I_1^*) = 1/R$, so $\tau'(I_1^*)R_b = R_b/R < 1$ and $\delta U_b = (1 - R_b/R)\, \delta r_l = \frac{R-R_b}{R}\, \delta r_l = (R_l/R)\, \delta r_l > 0$: the deal pays. The gain is exactly the loss $\tau'(I_1^*)R_l\, \delta r_l = (R_l/R)\delta r_l$ of the other long-term claimholders, whose success probability falls; total value is unchanged to first order because I_1^* maximises (69).

Step 4 — (ii) The indeterminacy is resolved. Lenders anticipate such deals, so only collusion-proof structures are equilibria. By (76) no deal pays iff

$$\tau'(I_1)\, R_b \geq 1. \tag{77}$$

The entrepreneur bears the NPV loss ex ante, and (69) rises with I_1 below I_1^*; since τ' falls with I_1, she sets I_1 as high as (77) allows, so (77) binds: $\tau'(I_1)R_b = 1$. Because $R_b < R$, $\tau'(I_1) > \tau'(I_1^*)$ and $I_1 < I_1^*$. A larger R_b permits a larger I_1, so R_b must be the largest compatible with breakeven, which requires paying her nothing at date 1, $r_b = 0$, and all date-1 free cash to lenders, $r_l = r - I_1$:

$$I = r - I_1 + [p + \tau(I_1)](R - R_b), \tag{78}$$
$$\tau'(I_1)\, R_b = 1. \tag{79}$$

These two equations pin down (I_1, R_b), hence $r_l = r - I_1$ and $R_l = R - R_b$. So yes, opacity resolves the indeterminacy: the entrepreneur is paid only long-term ($r_b = 0$), lenders hold more short-term debt ($r - I_1 > r - I_1^*$), and deepening investment is below the first best.

Answer (ii). At the first best a secret swap gives $\delta U_b = [1 - \tau'(I_1^*)R_b]\delta r_l = (R_l/R)\delta r_l > 0$. Yes, the indeterminacy is resolved: $r_b = 0$, $r_l = r - I_1$, and (I_1, R_b) solve $\tau'(I_1)R_b = 1$ and $I = r - I_1 + [p + \tau(I_1)](R - R_b)$, with $I_1 < I_1^*$ (a short-term bias).

Intuition. A lender who cashes out early is paid out of the firm's reinvestment fund, and the cost (a lower success probability) falls on every long-term claimant. Neither the colluding lender nor the entrepreneur internalises the loss of the other lenders, so with opaque books maturity is biased toward the short term. The entrepreneur can only commit not to strip the firm by holding a large long-term stake herself, and $\tau'(I_1)R_b = 1$ shows how large.

Common mistakes.

- Letting the colluding lender internalise the fall in I_1: he is atomistic, takes I_1 as given and accepts any deal with $\delta r_l \geq [p + \tau(I_1)]\delta R_l$.
- Counting the deal as value-creating: at I_1^* total value is unchanged to first order; the entrepreneur's gain is the other long-term holders' loss.
- Concluding that "more long-term debt" fixes the bias: what deters stripping is the entrepreneur's own long-term stake R_b, which is why $r_b = 0$.

Book answer. Matches the book's answer (p. 594): $\tau'(I_1^*)R = 1$; one degree of freedom; $\delta U_b = [1 - \tau'(I_1)R_b]\delta r_l > 0$; resolution with $r_b = 0$, $I = r - I_1 + [p + \tau(I_1)](R - R_b)$ and $\tau'(I_1)R_b = 1$, $I_1 < I_1^*$.

Ref: Tirole §5.6.1 "Optimal Claim Maturity", p. 226; Exercise 5.9, p. 233; answer p. 594.

10. One vs. two creditors: strategic and liquidity default — Amaro de Matos 5.2.5 #2, p. 148

What is asked. *(strategic)*: show that when the manager defaults strategically and buys the firm back, the creditors collect more if the assets are split between two creditors than if one creditor holds them all. *(liquidity)*: show that when the default is a genuine liquidity default and the assets are sold to an outside buyer, two creditors collect less in expectation than one.

Approach. This is Bolton and Scharfstein (1996) inside the strategic-default model of §5.2.1: the investor's recovery is $y - S$ after a strategic default and $L_\circ$ after a liquidity default, and both come from bargaining. With two creditors each holds an asset the other party needs to realise the synergy Δ, so each has hold-up power. Splitting the surplus by the Shapley value (the three-player version of the equal split $S = y/2$) gives both formulas on p. 138; comparing them with $y/2$ and $\alpha^2 y^2/(4\bar{c})$ answers the two parts.

Notation.

- K: investment; q: probability of cash flow x at $t = 1$ (else 0); y: $t = 2$ cash flow if the manager continues.
- S: the manager's payoff when she strategically defaults and buys the firm back; $y - S$: what the creditor(s) get.
- $\alpha \in [0, 1]$: efficiency of an outside buyer (αy); $c \sim U[0, \bar{c}]$: buyer's information cost.
- $L_\circ$ / $\overline{L}_\circ$: expected liquidation payoff to creditors after a zero cash flow, with one / two creditors.
- $y^A, y^B \geq 0$: cash flows of assets A and B run separately; $\Delta = y - y^A - y^B \in (0, y]$: synergy.
- $\beta_\circ$: liquidation probability after a zero report, $\beta_\circ = K/[q(y - S) + (1 - q)L_\circ]$ from (5.6).

Assumption. With two creditors, creditor 1 holds asset A and creditor 2 asset B. Multilateral bargaining follows the Shapley value, the three-player version of the equal split used for one creditor; disagreement payoffs are 0 (as behind $S = y/2$ in the one-creditor case). The book only cites Bolton and Scharfstein (1996), pp. 10–13; the derivation below is ours, and it reproduces both formulas printed on p. 138.

Step 1 — One creditor (benchmark). We first record the one-creditor recoveries the comparison is made against. Manager and creditor split y equally: $S = y/2$, so in a strategic default the creditor gets

$$y - S = \frac{y}{2}. \tag{80}$$

After a zero cash flow the buyer gets $\alpha y/2$, buys iff $c \le \alpha y/2$ (probability $\alpha y/(2\bar{c})$), and the creditor gets $\alpha y/2$:

$$L_\circ = \Big(\frac{\alpha y}{2\bar{c}}\Big)\Big(\frac{\alpha y}{2}\Big) = \frac{\alpha^2 y^2}{4\bar{c}}. \tag{81}$$

Step 2 — Two creditors, strategic default. Players: manager M, creditors 1 and 2. Worth of a coalition: y if it contains M and both assets, y^A (y^B) if it contains M and only asset A (B), and 0 without M. Over the $3! = 6$ orderings, M's marginal contribution is 0 when she comes first (orderings $M12$, $M21$), y^A in ordering $1M2$ and y^B in $2M1$ (she joins a coalition holding one asset, worth 0 without her), and y when last ($12M$, $21M$). Averaging over the six orderings and using $y^A + y^B = y - \Delta$:

$$\begin{aligned} S &= \frac{0 + 0 + y^A + y^B + y + y}{6} = \frac{2y + y^A + y^B}{6} \\ &= \frac{2y + (y - \Delta)}{6} = \frac{3y - \Delta}{6} \\ &= \frac{y}{2} - \frac{\Delta}{6}, \end{aligned} \tag{82}$$

which is the book's S. The two creditors together get

$$\begin{aligned} y - S &= y - \Big(\frac{y}{2} - \frac{\Delta}{6}\Big) \\ &= \frac{y}{2} + \frac{\Delta}{6} \;>\; \frac{y}{2}. \end{aligned} \tag{83}$$

Comparing with (80): two creditors are paid more in a strategic default, by $\Delta/6$. Each creditor can hold up the synergy Δ, so the manager keeps less when she buys the firm back.

Answer (strategic). Two creditors get $y - S = \dfrac{y}{2} + \dfrac{\Delta}{6} > \dfrac{y}{2}$ (one creditor): more, by $\Delta/6$.

Step 3 — Two creditors, liquidity default. Replace M by the outside buyer, whose coalition worths are αy, αy^A, αy^B (the cost c is sunk and not bargained over, as in Step 1). By (82) the buyer gets $\alpha(y/2 - \Delta/6)$, buys iff $c \le \alpha(y/2 - \Delta/6)$, which for $c \sim U[0, \bar{c}]$ has probability $\alpha(y/2 - \Delta/6)/\bar{c}$ (assume $\alpha(y/2 - \Delta/6) \le \bar{c}$), and the creditors share $\alpha y - \alpha(y/2 - \Delta/6) = \alpha(y/2 + \Delta/6)$. Expected liquidation payoff = probability of a sale times the creditors' share:

$$\begin{aligned} \overline{L}_\circ &= \Big(\frac{\alpha(3y - \Delta)}{6\bar{c}}\Big)\Big(\frac{\alpha(3y + \Delta)}{6}\Big) \\ &= \frac{\alpha^2(3y - \Delta)(3y + \Delta)}{36\bar{c}} = \frac{\alpha^2(9y^2 - \Delta^2)}{36\bar{c}} \\ &= \frac{9\alpha^2 y^2}{36\bar{c}} - \frac{\alpha^2\Delta^2}{36\bar{c}} = \frac{\alpha^2 y^2}{4\bar{c}} - \frac{\alpha^2\Delta^2}{36\bar{c}}, \end{aligned} \tag{84}$$

the book's formula. By (81), $\overline{L}_\circ = L_\circ - \alpha^2\Delta^2/(36\bar{c}) < L_\circ$: two creditors are paid less in a liquidity default. The creditors' extra share does not compensate the lower probability that a buyer shows up, which falls from $\alpha y/(2\bar{c})$ to $\alpha(y/2-\Delta/6)/\bar{c}$.

Answer (liquidity). Two creditors expect $\overline{L}_\circ = \dfrac{\alpha^2 y^2}{4\bar{c}} - \dfrac{\alpha^2\Delta^2}{36\bar{c}} < L_\circ = \dfrac{\alpha^2 y^2}{4\bar{c}}$: less, by $\alpha^2\Delta^2/(36\bar{c})$.

Step 4 — The $\beta_\circ$ denominator and the printed sign. By (5.6) the denominator of $\beta_\circ$ is $q(y-S)+(1-q)L$. With two creditors, substituting (83) and (84),

$$D_2 = q\Big(\frac{y}{2}+\frac{\Delta}{6}\Big) + (1-q)\Big(\frac{\alpha^2 y^2}{4\bar{c}} - \frac{\alpha^2\Delta^2}{36\bar{c}}\Big). \tag{85}$$

The book prints $q(\frac{y}{2}-\frac{\Delta}{6})$ in the first term (p. 138), i.e. qS instead of $q(y-S)$ (the two coincide only when $S=y/2$, the one-creditor case). That is a sign error: with the printed term the two-creditor denominator would be below the one-creditor one for every q, contradicting the same page's remarks that it "can be greater or less" and that two creditors are paid more in strategic default. With (85) and $D_1 = q\frac{y}{2} + (1-q)L_\circ$,

$$\begin{aligned} D_2 - D_1 &= q\Big(\frac{y}{2}+\frac{\Delta}{6}-\frac{y}{2}\Big) + (1-q)\big(\overline{L}_\circ - L_\circ\big) \\ &= q\,\frac{\Delta}{6} - (1-q)\,\frac{\alpha^2\Delta^2}{36\bar{c}}. \end{aligned} \tag{86}$$

It is positive (two creditors liquidate less often) iff, multiplying by $36\bar{c}/\Delta > 0$, $6\bar{c}\,q > (1-q)\alpha^2\Delta$, i.e. $q(6\bar{c}+\alpha^2\Delta) > \alpha^2\Delta$, i.e. $q > \bar{q} \equiv \frac{\alpha^2\Delta}{6\bar{c}+\alpha^2\Delta}$: when bankruptcy risk is low, as the book concludes.

Step 5 — Numerical illustration (ours). Take $y=10$, $y^A=4$, $y^B=3$ ($\Delta = 10-4-3=3$), $\alpha=0.8$, $\bar{c}=6$, $q=0.6$, $K=2$. Then $S = 10/2 - 3/6 = 5 - 0.5 = 4.5$ and $y-S = 5.5$ versus 5 with one creditor. In liquidity default the buyer gets $0.8(4.5)=3.6$, buys with probability $3.6/6 = 0.6$ (versus $4/6 = 0.666667$), and the creditors get $0.8(5.5) = 4.4$:

$$\overline{L}_\circ = (0.6)(4.4) = 2.64 \;<\; L_\circ = \frac{(0.64)(100)}{24} = 2.666667.$$

Denominators: $D_1 = (0.6)(5)+(0.4)(2.666667) = 4.066667$ and $D_2 = (0.6)(5.5)+(0.4)(2.64) = 4.356$, so $\beta_\circ$ falls from $2/4.066667 = 0.491803$ to $2/4.356 = 0.459137$. The misprinted term would give $(0.6)(4.5)+(0.4)(2.64) = 3.756$ and $\beta_\circ = 0.532481$. Here $\bar{q} = 1.92/37.92 = 0.050633$; at $q=0.02$, (86) gives $-0.016133 < 0$ and one creditor is better.

Intuition. Splitting the assets among creditors gives each a veto over the synergy Δ. This hold-up power makes strategic default costly for the manager, so more can be credibly repaid (good when default is mostly strategic, high q). The same hold-up power scares off outside buyers when the firm genuinely fails, lowering the liquidation value (bad when default is mostly for liquidity, low q).

Common mistakes.

- Comparing S instead of $y-S$: S is what the *manager* keeps; the creditors' recovery is $y-S$, which rises with two creditors.
- Keeping the buyer's probability of purchase fixed at $\alpha y/(2\bar{c})$: the buyer's share falls to $\alpha(y/2-\Delta/6)$, so fewer buyers show up, and that loss dominates.
- Copying the printed two-creditor denominator $q(y/2-\Delta/6)+\ldots$: its first term must be $q(y-S) = q(y/2+\Delta/6)$.

Book answer. The book (p. 268) only cites Bolton and Scharfstein (1996), pp. 10–13; derived here. Our S and $\overline{L}_\circ$ match p. 138. The printed two-creditor denominator on p. 138 has a sign error: its first term should be $q(y/2+\Delta/6) = q(y-S)$, not $q(y/2-\Delta/6)$.

Ref: Amaro de Matos §5.2.1 "Extending the Model", pp. 134–137; §5.2.2 "The Case of Many Creditholders", pp. 137–138; exercise p. 148.

11. When the good entrepreneur prefers liquidating debt — Amaro de Matos 5.2.5 #4, p. 148

What is asked. Prove that if $(q^d X + f^d C)/R \le L$, condition (5.12), then the entrepreneur of a good project prefers a debt structure that liquidates the firm after a downgrade to any structure that does not (the comparison of Lemma 5.2.2).

Approach. Diamond (1991): liquidation after a downgrade costs the good entrepreneur $X + C$ in that state but lets lenders recover L from the whole downgraded pool, most of which is bad, so the upgraded (good) firms pay less. Lemma 5.2.2 reduces the comparison to $C < \widehat{C}$. The proof therefore (1) derives both payoffs from the break-even (5.11) and Bayes' rule, (2) obtains $U^L - U^N = e(\widehat{C} - C)$, and (3) bounds $\widehat{C}$ from below using the fact that no non-liquidating debt can be worth more than $q^d X/R$ after a downgrade.

Notation.

- X: $t = 2$ payoff (good project surely, bad with probability π); $C \ge 0$: nonassignable control rent; L: liquidation value; R: gross return per period.
- f: initial rating; $f^u = 1$, f^d: posterior of "good" after an upgrade / downgrade; $e \in (0, 1)$: probability a good project is downgraded.
- P_d: probability of a downgrade for an unknown type; $q^d = \pi + f^d(1-\pi)$: repayment probability after a downgrade.
- V^d: $t = 1$ value of a non-liquidating debt after a downgrade; r_1: face value of short-term debt; U^N, U^L: good entrepreneur's payoffs under non-liquidating and liquidating debt.

Step 1 — Bayes. Both payoffs depend on the downgrade probability, which we link to f^d. Since $f^u = 1$, bad projects are always downgraded and good ones with probability e, so

$$P_d = fe + (1 - f) = 1 - (1 - e)f, \tag{87}$$

$$f^d = \frac{fe}{P_d} \quad \Longleftrightarrow \quad \frac{P_d}{f} = \frac{e}{f^d}. \tag{88}$$

Step 2 — Non-liquidating debt. We compute the good entrepreneur's payoff when a downgrade does not trigger liquidation. After a downgrade, the debt worth V^d at $t = 1$ requires a $t = 2$ payment RV^d/q^d (new or existing lenders break even at the pooled repayment probability q^d), which the good firm pays for sure; after an upgrade it pays $r_1 R$. (For long-term debt read $r_1 = \rho/R$: then the break-even $P_u(\rho/R) + P_d V_l^d = R$ is (5.11), since $P_u + P_d q^d = q$ and $\rho q = R^2$.) Rearranging (5.11), $(1 - e) f\, r_1 = R - P_d V^d$; multiplying by R/f and using (88),

$$\begin{aligned}(1 - e)\, r_1 R &= \frac{R\,(R - P_d V^d)}{f} \\ &= \frac{R^2}{f} - RV^d\,\frac{P_d}{f} \\ &= \frac{R^2}{f} - \frac{eRV^d}{f^d}.\end{aligned} \tag{89}$$

So, expanding and substituting (89),

$$\begin{aligned}U^N &= e\Big(X + C - \frac{RV^d}{q^d}\Big) + (1 - e)(X + C - r_1 R) \\ &= (X + C) - \frac{eRV^d}{q^d} - (1 - e)\, r_1 R \\ &= (X + C) - \frac{eRV^d}{q^d} - \frac{R^2}{f} + \frac{eRV^d}{f^d} \\ &= X + C - \frac{R^2}{f} + eRV^d\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big),\end{aligned} \tag{90}$$

the expression in the proof of Lemma 5.2.1.

Step 3 — Liquidating debt. Now a downgraded firm is liquidated and the lenders get L, so (89) holds with $V^d = L$, and the good entrepreneur gets nothing after a downgrade:

$$\begin{aligned} U^L &= (1-e)(X+C) - (1-e)\, r_1 R \\ &= (1-e)(X+C) - \Big(\frac{R^2}{f} - \frac{eRL}{f^d}\Big) \\ &= (1-e)(X+C) - \frac{R^2}{f} + \frac{eRL}{f^d}. \end{aligned} \tag{91}$$

Step 4 — Lemma 5.2.2. We now compare the two. Subtracting (90) from (91), the R^2/f terms cancel and $(1-e)(X+C) - (X+C) = -e(X+C)$:

$$\begin{aligned} U^L - U^N &= -e(X+C) + \frac{eRL}{f^d} - eRV^d\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big) \\ &= e\Big[\frac{LR}{f^d} - X - RV^d\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big) - C\Big] \\ &= e\,(\widehat{C} - C), \end{aligned} \tag{92}$$

so (with $e > 0$) the good entrepreneur prefers liquidation iff $C < \widehat{C}$.

Step 5 — A bound on V^d. A downgraded firm can pay at most X at $t = 2$, and repays with probability q^d; so any non-liquidating debt has

$$V^d \le \frac{q^d X}{R}. \tag{93}$$

(Long-term: $V_l^d = q^d \rho / R$ with $\rho \le X$. Short-term without liquidation: $V_s^d = r_1 \le q^d X/R$, the refinancing condition.)

Step 6 — (5.12) is sufficient. Since $\frac{1}{f^d} - \frac{1}{q^d} > 0$ by (5.10), $\widehat{C}$ falls with V^d; substituting (93) into the definition of $\widehat{C}$,

$$\begin{aligned} \widehat{C} &\ge \frac{LR}{f^d} - X - R\Big(\frac{q^d X}{R}\Big)\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big) \\ &= \frac{LR}{f^d} - X - \frac{q^d X}{f^d} + \frac{q^d X}{q^d} \\ &= \frac{LR}{f^d} - X - \frac{q^d X}{f^d} + X \\ &= \frac{LR - q^d X}{f^d}. \end{aligned} \tag{94}$$

Inequality (5.12), $(q^d X + f^d C)/R \le L$, i.e. $q^d X + f^d C \le LR$, is equivalent (subtract $q^d X$, divide by $f^d > 0$) to

$$C \le \frac{LR - q^d X}{f^d}. \tag{95}$$

Chaining (95) and (94): $C \le \widehat{C}$, and by (92) $U^L \ge U^N$. The preference is strict unless both (95) and (93) hold with equality (a knife-edge with $\rho = X$ or $r_1 = q^d X/R$). This proves the claim.

Answer (proof). For any non-liquidating structure, $V^d \le q^d X/R$ gives $\widehat{C} \ge (LR - q^d X)/f^d$; (5.12) is $C \le (LR - q^d X)/f^d$; hence $C \le \widehat{C}$ and $U^L - U^N = e(\widehat{C} - C) \ge 0$: the good entrepreneur prefers the liquidating structure.

Step 7 — Remark on $\widehat{C} > 0$. (94) also shows $\widehat{C} \geq 0$ whenever $LR \geq q^d X$, i.e. whenever lenders really prefer liquidating a downgraded firm to rolling it over — the condition for a liquidating structure to exist. Without it, $\widehat{C}$ can be negative: with the numbers of Amaro de Matos exercise 5.2.5 #5 ($L = R = 1$, $X = \frac{7}{4}$, $f^d = \frac{1}{3}$, $q^d = \frac{2}{3}$) and short-term debt ($V^d = 1$), $\widehat{C} = 3 - 1.75 - (1)(1)(3 - \frac{3}{2}) = -0.25$; so the lemma's "$> 0$" presumes credible liquidation.

Intuition. Liquidation costs the good entrepreneur her firm (and C) in the downgrade state, but it shifts financing costs onto the pool of downgraded firms, most of whom are bad: lenders recover L from every downgraded firm, so the upgraded, i.e. good, firms pay less. (5.12) says the continuation value of a downgraded firm — what lenders could ever collect, $q^d X$, plus the control rent weighted by the share of good types, $f^d C$ — is below L; then the cross-subsidy gain outweighs the lost rent.

Common mistakes.

- Forgetting that a liquidated good entrepreneur loses both X and the control rent C in the downgrade state: U^L has $(1 - e)(X + C)$, not $X + C$.
- Signing $\frac{1}{f^d} - \frac{1}{q^d}$ wrongly: $q^d > f^d$ by (5.10), so it is positive and $\widehat{C}$ *decreases* in V^d; the bound $V^d \leq q^d X/R$ is therefore the worst case for liquidation.
- Dropping R when moving between (5.12) and $C \leq (LR - q^d X)/f^d$: (5.12) is stated per unit of R.

Book answer. The book (p. 268) only cites the appendix of Diamond (1991), pp. 732–733; proved here.

Ref: Amaro de Matos §5.2.3 "The Choice of the Duration", pp. 139–142 (Lemmas 5.2.1–5.2.2, (5.9)–(5.12)); exercise p. 148.

12. Diamond (1991): liquidation after a downgrade — Amaro de Matos 5.2.5 #6 (with #5), p. 148

What is asked. *Part 0* (exercise #5, needed as the base case): (a) face values of short- and long-term debt; (b) show there is no liquidation; (c) the good entrepreneur's payoff with long-term debt; (d) the probability e that a good project is downgraded; (e) her payoff with short-term debt. *(i)* With $X = 1.4$, $C = 0.35$, show that short-term debt now leads to liquidation after a downgrade. *(ii)* Show that the good entrepreneur then prefers long-term debt. *(iii)* Show that this preference reverses when the initial rating is $f = \frac{3}{4}$.

Approach. Diamond's (1991) maturity trade-off. Short-term debt is repriced at $t = 1$ on nonverifiable rating news: without liquidation that only shifts repayment onto the downgraded pool, which helps good types (Lemma 5.2.1); with liquidation it also costs the good type $X + C$ whenever she is downgraded (Lemma 5.2.2: liquidating debt wins iff $C < \widehat{C}$). A downgraded borrower is liquidated when the most she can promise, $q^d X/R$, is below both r_1 and L. The rating f matters through e: a higher f (with f^d fixed) means good types are rarely downgraded, so the liquidation cost is rarely paid.

Notation.

- X: $t = 2$ payoff (good project surely, bad with probability π); $C \geq 0$: control rent; L: liquidation value; R: gross return per period.
- f: initial rating; $f^u = 1$, f^d: posterior of "good" after an upgrade / downgrade; e: probability that a good project is downgraded; P_d: probability of a downgrade for an unknown type; $q^d = \pi + f^d(1 - \pi)$; V^d: $t = 1$ value of the debt after a downgrade (V_s^d short, V_l^d long).
- ρ: face value of long-term debt; r_1: face value of short-term debt; q: $t = 0$ repayment probability (5.9); F^d: face value of the $t = 1$ refinancing after a downgrade; U_{LT}, U_{ST}: good entrepreneur's payoff with long- / short-term debt.

- Part 0 (#5): $L = R = 1$, $\pi = \frac{1}{2}$, $X = \frac{7}{4}$, $C = 0$, $f = \frac{1}{2}$, $f^d = \frac{1}{3}$. Exercise #6: $X = 1.4$, $C = 0.35$, then $f = \frac{3}{4}$.

Assumption. (a) $L = 1 = K$ contradicts the text's $L < K$; it is harmless here but makes short-term debt riskless ($r_1 = R$). (b) When f becomes $\frac{3}{4}$ we keep $f^d = \frac{1}{3}$, as the statement says ("keep everything") and as Proposition 5.2.1 does; then e adjusts. The book only cites Diamond (1991), examples 1–3; all numbers below are derived here.

Step 1 — Part 0 (a): face values. Each face value is set by the lenders' break-even at its horizon. By (5.9),

$$q = \pi + f(1-\pi) = \tfrac{1}{2} + (\tfrac{1}{2})(\tfrac{1}{2}) = \tfrac{3}{4}. \tag{96}$$

Long-term debt satisfies $\rho q = R^2$:

$$\rho = \frac{R^2}{q} = \frac{(1)}{(3/4)} = \frac{4}{3} \ \leq X. \tag{97}$$

For short-term debt we need e, P_d, q^d. Bad projects are always downgraded ($f^u = 1$) and good ones with probability e, so $P_d = fe + (1-f)$ and Bayes' rule gives

$$f^d = \frac{fe}{fe + (1-f)}. \tag{98}$$

Solving for e: $f^d[fe + (1-f)] = fe$, so $f^d(1-f) = fe(1-f^d)$ and

$$e = \frac{f^d(1-f)}{f(1-f^d)} = \frac{(1/3)(1/2)}{(1/2)(2/3)} = \frac{1/6}{1/3} = \frac{1}{2}, \tag{99}$$

$$P_d = 1 - (1-e)f = 1 - (\tfrac{1}{2})(\tfrac{1}{2}) = \tfrac{3}{4}, \tag{100}$$

$$q^d = \pi + f^d(1-\pi) = \tfrac{1}{2} + (\tfrac{1}{3})(\tfrac{1}{2}) = \tfrac{2}{3}. \tag{101}$$

A downgraded firm can roll over up to $q^d X/R = (2/3)(7/4) = 7/6 > L$. Conjecture no liquidation, $V_s^d = r_1$; then (5.11), multiplied by $(1-e)f$, gives

$$\begin{aligned} (1-e)f\, r_1 &= R - P_d\, r_1, \\ r_1\big[(1-e)f + P_d\big] &= R, \\ r_1 &= R = 1, \end{aligned} \tag{102}$$

since $(1-e)f + P_d = 1$. Indeed $r_1 = 1 \leq 7/6$, so $V_s^d = \min[1, \max(1, 7/6)] = 1 = r_1$: consistent.

Answer 0(a). $r_1 = 1$, $\rho = \frac{4}{3}$.

Step 2 — Part 0 (b): no liquidation. Long-term debt has no $t = 1$ payment. A downgraded short-term borrower refinances $r_1 = 1$ with face

$$F^d = \frac{Rr_1}{q^d} = \frac{(1)(1)}{(2/3)} = \frac{3}{2} \ \leq X = \frac{7}{4}, \tag{103}$$

and the lenders would not liquidate anyway since $q^d X/R = 7/6 > L = 1$.

Answer 0(b). No liquidation: the downgraded borrower refinances at face $F^d = \frac{3}{2} \leq X$.

Step 3 — Part 0 (c)–(e). (c) The good project always pays ρ: $U_{LT} = X + C - \rho = \frac{7}{4} - \frac{4}{3} = \frac{21-16}{12} = \frac{5}{12} = 0.416667$. (d) By (99), $e = \frac{1}{2}$. (e) Upgraded firms roll over at $r_1 R = 1$, downgraded ones pay F^d:

$$\begin{aligned} U_{ST} &= e(X + C - F^d) + (1-e)(X + C - r_1 R) \\ &= (\tfrac{1}{2})(\tfrac{7}{4} - \tfrac{3}{2}) + (\tfrac{1}{2})(\tfrac{7}{4} - 1) \\ &= (\tfrac{1}{2})(\tfrac{1}{4}) + (\tfrac{1}{2})(\tfrac{3}{4}) = \tfrac{1}{8} + \tfrac{3}{8} = \tfrac{1}{2}. \end{aligned} \tag{104}$$

Short-term debt is preferred ($\frac{1}{2} > \frac{5}{12}$), as Lemma 5.2.1 predicts: by the Lemma's payoff expression, with $R^2/f = 2$ and $e(\frac{1}{f^d} - \frac{1}{q^d}) = \frac{1}{2}(3 - \frac{3}{2}) = \frac{3}{4}$,

$$U = X + C - \frac{R^2}{f} + eRV^d\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big) = \tfrac{7}{4} + 0 - 2 + \tfrac{3}{4}V^d = -\tfrac{1}{4} + \tfrac{3}{4}V^d, \tag{105}$$

with $V_s^d = 1$ (giving $-\frac{1}{4} + \frac{3}{4} = \frac{1}{2}$) $> V_l^d = q^d\rho/R = \frac{8}{9}$ (giving $-\frac{1}{4} + \frac{2}{3} = \frac{5}{12}$).

Answer 0(c)–(e). (c) $U_{LT} = \frac{5}{12} \approx 0.416667$; (d) $e = \frac{1}{2}$; (e) $U_{ST} = \frac{1}{2}$.

Step 4 — #6, $f = \frac{1}{2}$: short-term debt is liquidated after a downgrade. Now $X = 1.4$, $C = 0.35$; q, ρ, e, P_d, q^d are unchanged, and $\rho = \frac{4}{3} \le 1.4$ keeps long-term debt feasible. A downgraded firm can now roll over at most

$$\frac{q^d X}{R} = (\tfrac{2}{3})(1.4) = \tfrac{14}{15} = 0.933333 \;<\; L = 1. \tag{106}$$

If it were not liquidated, (102) would give $r_1 = 1 > 14/15$: it could not refinance. So it defaults, and the lenders, offered at most $14/15$ by continuation against $L = 1$ from liquidation, liquidate. Hence $V_s^d = \max(L, q^d X/R) = 1$ and by (5.11)

$$r_1 = \frac{R - P_d V_s^d}{(1-e)f} = \frac{1 - (3/4)(1)}{(1/4)} = 1, \tag{107}$$

consistent with $V_s^d = \min[1, \max(1, 14/15)] = 1$.

Answer (i). $q^d X/R = \frac{14}{15} < r_1 = 1$ and $< L = 1$: a downgraded short-term borrower cannot refinance and is liquidated.

Step 5 — #6, $f = \frac{1}{2}$: long-term debt is preferred. The good firm keeps control only after an upgrade:

$$U_{ST} = (1-e)(X + C - r_1 R) = (\tfrac{1}{2})(1.4 + 0.35 - 1) = (\tfrac{1}{2})(0.75) = 0.375, \tag{108}$$

$$U_{LT} = X + C - \rho = 1.75 - \tfrac{4}{3} = 0.416667. \tag{109}$$

$U_{LT} > U_{ST}$. To check this against Lemma 5.2.2, write both payoffs with (5.11) and Bayes ($P_d/f = e/f^d$, from (98)). The liquidating short-term debt has $V_s^d = L$, so $(1-e)r_1 R = R(R - P_d L)/f = R^2/f - eRL/f^d$ and $U_{ST} = (1-e)(X+C) - R^2/f + eRL/f^d$; the long-term debt has the Lemma 5.2.1 payoff $U_{LT} = X + C - R^2/f + eRV_l^d(1/f^d - 1/q^d)$. Subtracting,

$$U_{ST} - U_{LT} = -e(X + C) + \frac{eRL}{f^d} - eRV_l^d\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big) = e(\widehat{C} - C), \tag{110}$$

with $\widehat{C}$ as in Lemma 5.2.2. With $V^d = V_l^d = \frac{8}{9}$: $\widehat{C} = \frac{(1)(1)}{1/3} - 1.4 - (1)(\frac{8}{9})(3 - \frac{3}{2}) = 1.6 - \frac{4}{3} = 0.266667 < C = 0.35$, and (110) gives $U_{ST} - U_{LT} = (0.5)(0.266667 - 0.35) = -0.041667$. (Condition (5.12) fails: $(q^d X + f^d C)/R = 0.933333 + 0.116667 = 1.05 > L$.)

Answer (ii). $U_{LT} = 0.416667 > U_{ST} = 0.375$ (gap $-0.041667 = e(\widehat{C} - C)$ with $\widehat{C} = 0.266667 < C$): long-term debt is preferred.

Step 6 — #6, $f = \frac{3}{4}$: the conclusion reverses. With $f^d = \frac{1}{3}$ kept, (99)–(100) and (96)–(97) give

$$e = \frac{(1/3)(1/4)}{(3/4)(2/3)} = \frac{1/12}{1/2} = \frac{1}{6}, \quad P_d = 1 - (\tfrac{5}{6})(\tfrac{3}{4}) = 1 - \tfrac{5}{8} = \tfrac{3}{8},$$

$$q = \tfrac{1}{2} + (\tfrac{3}{4})(\tfrac{1}{2}) = \tfrac{7}{8}, \quad \rho = \frac{(1)}{(7/8)} = \tfrac{8}{7} = 1.142857.$$

$q^d = \frac{2}{3}$ is unchanged, so (106) still holds and short-term debt is still liquidated after a downgrade; with $(1-e)f = (\frac{5}{6})(\frac{3}{4}) = \frac{5}{8}$, (5.11) gives $r_1 = \frac{1-(3/8)(1)}{(5/8)} = \frac{5/8}{5/8} = 1$. Then

$$U_{ST} = (1-e)(X + C - r_1 R) = (\tfrac{5}{6})(0.75) = 0.625, \tag{111}$$

$$U_{LT} = X + C - \rho = 1.75 - \tfrac{8}{7} = 0.607143, \tag{112}$$

so now short-term (liquidating) debt is preferred. Check: $V_l^d = q^d\rho/R = (\frac{2}{3})(\frac{8}{7}) = \frac{16}{21}$, $\widehat{C} = 1.6 - (\frac{16}{21})(\frac{3}{2}) = 1.6 - \frac{8}{7} = 0.457143 > C = 0.35$, and by (110) $e(\widehat{C} - C) = \frac{1}{6}(0.107143) = 0.017857 = U_{ST} - U_{LT}$. Proposition 5.2.1 (p. 142) agrees: its bound is $\frac{\pi[R^2 - LR + f^d(C+X-R^2)]}{(1-\pi)[LR - f^d(C+X)]} = \frac{(1/2)[1-1+(1/3)(0.75)]}{(1/2)[1-(1/3)(1.75)]} = \frac{(1/2)(0.25)}{(1/2)(0.416667)} = 0.6$, and $f = \frac{1}{2} < 0.6 < \frac{3}{4}$.

Answer (iii). At $f = \frac{3}{4}$ ($e = \frac{1}{6}$): $U_{ST} = 0.625 > U_{LT} = 0.607143$ (gap 0.017857): short-term, liquidating debt is now preferred.

Intuition. With a low rating the good firm is downgraded often ($e = \frac{1}{2}$); liquidation then costs it the control rent too often, and long-term debt, which never triggers liquidation, wins. With a high rating, good firms are rarely downgraded ($e = \frac{1}{6}$), while the downgraded pool is mostly bad firms; liquidating that pool lets good firms borrow cheaply after the upgrade, which now outweighs the rare loss of control. (If instead $e = \frac{1}{2}$ were kept at $f = \frac{3}{4}$, then $f^d = \frac{3}{5}$, $q^d X/R = 1.12 > L$, there is no liquidation, and short-term debt still wins by Lemma 5.2.1, 0.625 vs 0.607143.)

Common mistakes.

- Keeping $e = \frac{1}{2}$ when f rises to $\frac{3}{4}$: the statement keeps everything else, including $f^d = \frac{1}{3}$, so e must be recomputed by Bayes ($e = \frac{1}{6}$).
- Giving the liquidated good entrepreneur something after a downgrade: liquidation removes X and C, so $U_{ST} = (1-e)(X + C - r_1 R)$ only.
- Checking only $q^d X/R < r_1$ for liquidation: lenders liquidate because, in addition, L beats the best refinancing offer, $L = 1 > q^d X/R = \frac{14}{15}$.

Book answer. The book (p. 268) only cites examples 1–3 of Diamond (1991), pp. 716–719; derived here. The three claims of #6 are confirmed (liquidation after downgrade; long-term preferred at $f = \frac{1}{2}$; reversal at $f = \frac{3}{4}$), consistent with Proposition 5.2.1.

Ref: Amaro de Matos §5.2.3 “The Choice of the Duration”, pp. 139–142 ((5.9)–(5.12), Lemmas 5.2.1–5.2.2, Proposition 5.2.1); exercises p. 148.

13. A borrower-friendly bankruptcy court — Tirole Exercise 4.9, pp. 190–191

What is asked. (i) Explain why the date-1 profit r and the liquidation value L should go to investors, why the borrower's stake on continuation is exactly $B/\Delta p$, and what conditions (1) and (2) mean. (ii) Write the borrower's optimisation over the continuation rule and give the equation for the cutoff r^*. (iii) Show that short-term debt $d = r^*$ implements the optimum (liquidation = bankruptcy) and say how d moves with the borrower's equity A. (iv) If a court gives the borrower half of L in bankruptcy and $r^* > \hat{r}$, show that the borrower can no longer be financed. (v) If $r^* < \hat{r}$, show that the court either blocks financing or makes bankruptcy more likely, hurting the borrower and not the lenders.

Approach. Tirole's fixed-investment model with an interim, verifiable signal r and a liquidation option. Continuation is always efficient but pledges only $\rho_0(r) < \rho_1(r)$, while liquidation pledges all of L; when pledgeable income is scarce, the optimal contract buys it with a *threat of termination* after bad news, which short-term debt implements (default ⇒ creditors take control and liquidate). Anything that lowers what investors get in bankruptcy lowers pledgeable income ex ante; competitive lenders still break even, so the cost falls on the borrower.

Notation.

- I, $A < I$: investment and cash; $r \in [0, \bar{r}]$: verifiable date-1 profit, density f, c.d.f. F, mean $E[r]$.
- $L > 0$: liquidation value; R: date-2 success profit; $p_H(r)$, $p_L(r)$: success probabilities (work / shirk), $p'_H > 0$, Δp constant; B: private benefit.
- $\rho_1(r) = p_H(r)R$, $\rho_0(r) = p_H(r)[R - B/\Delta p]$: expected and pledgeable continuation income, both increasing in r.
- r^*: continuation cutoff; d: short-term debt; $\hat{r}$: $\rho_0(\hat{r}) = L/2$; r^{**}: cutoff under the borrower-friendly court.
- $\mathcal{P}(x)$: investors' expected income with cutoff x; NPV(x): total surplus with cutoff x.

Step 1 — (i) Who gets r and L; $R_b = B/\Delta p$; conditions (1)–(2). r and L are verifiable and not subject to moral hazard, so pledging them to investors costs no incentives. Any part of them left to the borrower lowers pledgeable income, which (Step 2) must then be recovered by liquidating more often, an efficiency loss by (1). For the same reason, in continuation the incentive constraint $[p_H(r) - p_L(r)]R_b \geq B$, i.e. $(\Delta p)R_b \geq B$, should bind: $R_b = B/\Delta p$. Allowing $R_b(r) \geq B/\Delta p$ in the program of Step 2 leaves the NPV unchanged (a transfer) but lowers the left side of the breakeven constraint by $\int_{r^*}^{\bar{r}} p_H(r)[R_b(r) - B/\Delta p]f\,dr$, forcing a higher r^*.

Interpretation. (1) $\rho_1(r) > L$: continuation is always efficient. (2), right inequality $I - A > E[r + \rho_0(r)]$: always continuing leaves too little pledgeable income to finance the project. (2), left inequality $E[r] + L > I - A$: always liquidating would raise enough. So financing requires some inefficient liquidation.

Answer (i). Give r and L to investors (they cost no incentives, and anything left to the borrower must be recovered by liquidating more often); $R_b = B/\Delta p$ on continuation. (1): continuation is always efficient; (2): the project cannot be financed without some liquidation, but can be with enough of it.

Step 2 — (ii) Program and cutoff. With cutoff x (continue iff $r \geq x$), the borrower maximises the NPV (lenders break even, so she receives the whole surplus): date-1 profit, plus $\rho_1(r)$ on continuation states, plus L on liquidation states (probability $F(x)$):

$$\max_x \ \text{NPV}(x) = E[r] + \int_x^{\bar{r}} \rho_1(r)f(r)\,dr + F(x)L - I \tag{113}$$

$$\text{s.t.} \quad \mathcal{P}(x) \equiv E[r] + \int_x^{\bar{r}} \rho_0(r)f(r)\,dr + F(x)L \ \geq \ I - A. \tag{114}$$

(A cutoff rule is without loss: continuing at r gains $\rho_1(r) - L > 0$ in value and costs $L - \rho_0(r)$ in pledgeable income; the gain/cost ratio rises with r, so continuation goes to the highest r's.) Differentiating by Leibniz's rule, $\frac{d}{dx}\int_x^{\bar{r}} g(r)f(r)\,dr = -g(x)f(x)$ and $\frac{d}{dx}F(x)L = f(x)L$, so

$$\begin{aligned} \text{NPV}'(x) &= -\rho_1(x)f(x) + f(x)L \\ &= -f(x)[\rho_1(x) - L] < 0, \end{aligned} \tag{115}$$

$$\mathcal{P}'(x) = f(x)[L - \rho_0(x)]. \tag{116}$$

By (115) the borrower wants the lowest x satisfying (114), so r^* is the lowest root of

$$E[r] + \int_{r^*}^{\bar{r}} \rho_0(r)f(r)\,dr + F(r^*)L = I - A. \tag{117}$$

By (2), $\mathcal{P}(0) = E[r + \rho_0(r)] < I - A < E[r] + L = \mathcal{P}(\bar{r})$, so $r^* \in (0, \bar{r})$; since ρ_0 is increasing, $\mathcal{P}$ rises while $\rho_0 < L$ and falls after, so at the lowest root $\rho_0(r^*) \leq L$. (The statement's "$r^* \in (0, r)$" should read $(0, \bar{r})$.)

Answer (ii). $\max_x E[r] + \int_x^{\bar r} \rho_1 f\, dr + F(x)L - I$ s.t. $E[r] + \int_x^{\bar r} \rho_0 f\, dr + F(x)L \geq I - A$; r^* is the lowest root of $E[r] + \int_{r^*}^{\bar r} \rho_0(r) f(r)\, dr + F(r^*)L = I - A$.

Step 3 — (iii) Short-term debt and equity. Let the firm owe short-term debt $d = r^*$ at date 1 (plus, inter alia, the residual $r - r^*$ to investors and a long-term claim $R - B/\Delta p$ on success). If $r \geq r^*$, it pays d out of r and continues. If $r < r^*$, it defaults; creditors take control and liquidate ("bankruptcy"), collecting $r + L$. They do not renegotiate, because continuing would give them at most $\rho_0(r) < \rho_0(r^*) \leq L$. Default cannot be strategic since r is verifiable. Differentiating (117) with respect to A and using (116),

$$f(r^*)\left[L - \rho_0(r^*)\right] \frac{dr^*}{dA} = -1$$
$$\frac{dr^*}{dA} = -\frac{1}{f(r^*)\left[L - \rho_0(r^*)\right]} < 0. \tag{118}$$

Less equity requires more pledgeable income, obtained by liquidating more often: short-term debt rises.

Answer (iii). Short-term debt $d = r^*$: the firm continues iff it can pay d; otherwise creditors take control and liquidate. $dd/dA = dr^*/dA = -1/\{f(r^*)[L - \rho_0(r^*)]\} < 0$: less equity, more short-term debt.

Step 4 — (iv) Court splits L, case $r^* > \hat r$. Investors now get $L/2$ on liquidation. Whatever the continuation policy, their income at state r (besides r) is at most $\max\{L/2, \rho_0(r)\}$, so the best they can be promised is

$$\mathcal{P}^c_{\max} = E[r] + \int_0^{\bar r} \max\{L/2, \rho_0(r)\} f(r)\, dr. \tag{119}$$

Compare state by state with the contract of (ii), which gives L on $[0, r^*)$ and $\rho_0(r)$ on $[r^*, \bar r]$. For $r < \hat r$: $\max = L/2 < L$. For $\hat r \leq r < r^*$: $\max = \rho_0(r) < \rho_0(r^*) \leq L$. For $r \geq r^* > \hat r$: $\max = \rho_0(r)$, the same. Hence $\mathcal{P}^c_{\max} < \mathcal{P}(r^*) = I - A$ by (117): no contract lets investors break even, and the borrower loses access to financing.

Answer (iv). With $r^* > \hat r$, even the best policy under the court yields $\mathcal{P}^c_{\max} < I - A$: no financing.

Step 5 — (v) Case $r^* < \hat r$. With cutoff x, investors get $\mathcal{P}^c(x) = E[r] + \int_x^{\bar r} \rho_0 f\, dr + F(x)L/2$, with $\mathcal{P}^{c\prime}(x) = f(x)[L/2 - \rho_0(x)]$, positive below $\hat r$ and negative above. At the old cutoff,

$$\mathcal{P}^c(r^*) = \mathcal{P}(r^*) - F(r^*)\frac{L}{2} = I - A - F(r^*)\frac{L}{2} < I - A. \tag{120}$$

So if $\mathcal{P}^c(\hat r) < I - A$ there is no financing; otherwise the new cutoff $r^{**} \in (r^*, \hat r]$ solves $\mathcal{P}^c(r^{**}) = I - A$: bankruptcy becomes more likely. Lenders break even in every case (or do not lend), so they are not hurt. The borrower receives the NPV, and by (115) $\text{NPV}(r^{**}) < \text{NPV}(r^*)$: she is hurt (or loses financing altogether). Revealed preference says the same: under an enforcing court she could have promised investors only $L/2$ in bankruptcy, and chose not to.

Answer (v). Either $\mathcal{P}^c(\hat r) < I - A$ (no financing) or the cutoff rises to $r^{**} \in (r^*, \hat r]$ (more bankruptcy); lenders always break even, the borrower's NPV falls.

Step 6 — Numerical illustration (ours). $r \sim U[0,1]$ ($E[r] = 0.5$), $p_H(r) = 0.3 + 0.5r$, $R = 2$, $B/\Delta p = 1.6$ (e.g. $B = 0.4$, $\Delta p = 0.25$), so $\rho_1(r) = (0.3 + 0.5r)(2) = 0.6 + r$, $\rho_0(r) = (0.3 + 0.5r)(2 - 1.6) = 0.12 + 0.2r$, and $E[r + \rho_0] = 0.5 + 0.12 + 0.2(0.5) = 0.72$. With $f = 1$ on $[0,1]$,

$$\mathcal{P}(x) = 0.5 + \int_x^1 (0.12 + 0.2r)\, dr + Lx = 0.5 + 0.12(1 - x) + 0.1(1 - x^2) + Lx$$
$$= 0.72 + (L - 0.12)x - 0.1x^2.$$

Case (v): $L = 0.5$, $I - A = 0.75$ (conditions (1)–(2) hold: $0.6 > 0.5$, $1 > 0.75 > 0.72$). (117) is $0.72 + 0.38x - 0.1x^2 = 0.75$, i.e. $0.1x^2 - 0.38x + 0.03 = 0$, with discriminant $0.38^2 - 4(0.1)(0.03) = 0.1444 - 0.012 = 0.1324$; the lowest root is

$$r^* = \frac{0.38 - \sqrt{0.1324}}{0.2} = \frac{0.38 - 0.363868}{0.2} = 0.080659,$$

with $\rho_0(r^*) = 0.12 + 0.2(0.080659) = 0.136132$ and, by (118) with $f = 1$, $dr^*/dA = -1/(0.5 - 0.136132) = -1/0.363868 = -2.748249$. Here $\hat{r}$ solves $0.12 + 0.2\hat{r} = 0.25$, so $\hat{r} = (0.25 - 0.12)/0.2 = 0.65 > r^*$. Under the court $\mathcal{P}^c(x) = 0.72 + (0.25 - 0.12)x - 0.1x^2 = 0.72 + 0.13x - 0.1x^2$, so $\mathcal{P}^c(\hat{r}) = 0.72 + 0.0845 - 0.04225 = 0.76225 \geq 0.75$, and $\mathcal{P}^c(r^{**}) = 0.75$ gives $0.1x^2 - 0.13x + 0.03 = 0$, discriminant $0.0169 - 0.012 = 0.0049$, $r^{**} = (0.13 - \sqrt{0.0049})/0.2 = (0.13 - 0.07)/0.2 = 0.3$. The borrower's loss is $\int_{r^*}^{r^{**}} (\rho_1 - L)\, dr = \int_{r^*}^{r^{**}} (0.1 + r)\, dr = 0.1(0.3 - 0.080659) + (0.09 - 0.080659^2)/2 = 0.021934 + 0.041747 = 0.063681$. *Case (iv):* $L = 0.3$, $I - A = 0.78$: (117) is $0.1x^2 - 0.18x + 0.06 = 0$, discriminant $0.0324 - 0.024 = 0.0084$, so $r^* = (0.18 - 0.091652)/0.2 = 0.441742$, with $\rho_0(r^*) = 0.208348 \leq L$; $\hat{r} = (0.15 - 0.12)/0.2 = 0.15 < r^*$. (119) gives $0.5 + \int_0^{0.15} 0.15\, dr + \int_{0.15}^1 (0.12 + 0.2r)\, dr = 0.5 + (0.15)(0.15) + 0.12(0.85) + 0.1(1 - 0.0225) = 0.5 + 0.0225 + 0.102 + 0.09775 = 0.72225 < 0.78$: no financing.

Intuition. Bankruptcy is the borrower's commitment device: handing L to investors after bad news is what makes the loan pay. A court that gives the borrower half of L ex post destroys pledgeable income ex ante. Competitive lenders still break even, so the whole cost lands on the borrower: more frequent (inefficient) bankruptcy, or no loan at all.

Common mistakes.

- Taking the highest root of the break-even equation: liquidation is inefficient by (1), so the borrower wants the *lowest* cutoff that still lets investors break even.
- Concluding that the court helps borrowers or hurts lenders: lenders anticipate the split and break even (or do not lend); the whole cost is borne by the borrower.
- Thinking the default could be strategic here: r is verifiable, so default happens only when $r < d$ and creditors liquidate because $\rho_0(r) < L$.

Book answer. Matches the book's answer (pp. 586–587): same program, r^* the lowest value satisfying breakeven, $d = r^*$, $dr^*/dA < 0$, no financing when $r^* > \hat{r}$, and when $r^* < \hat{r}$ either no financing or bankruptcy over a larger interval $[0, r^{**})$, $r^{**} > r^*$, hurting the borrower only. (The book's wording "no bankruptcy occurs on some interval $[0, r^{**}]$" must mean bankruptcy occurs there.)

Ref: Tirole §3.2.1 "The Fixed-Investment Model" and §3.2.2 "The Lenders' Credit Analysis", pp. 115–117; Exercise 4.9, pp. 190–191; answer pp. 586–587.